Digital EGRA/EGMA Work Order No 4 Deliverable 4: Results of Pilot Test in Kiswahili

Including Test-Retest of the Pilot Self-Administered Early Grade Reading Assessment (SA-EGRA) and Self-Administered Early Grade Mathematics Assessment (SA-EGMA) and Concurrent Validity with Traditional EGRA/EGMA

Prepared for

Dr. Karen Levesque, Head of Research
Imagine Worldwide
1080 Edgewood Ave
Mill Valley, CA 94941Address
karen.levesque@imagineworldwide.org

Prepared by

Carmen Strigel, International Education
RTI International
3040 E. Cornwallis Road
Durham, NC 27713-2852
www.rti.org

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Carmen Strigel
Sr. Director, Center for Advanced Analytics and Visualization
RTI International
3040 E. Cornwallis Road
Durham, NC 27713-2852
www.rti.org

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Executive Summary

This report summarizes the findings of an effort to develop and validate a tablet-based, self-administered assessment of Kiswahili-language foundational literacy and numeracy in the early grades in Tanzania. RTI International developed the two assessments, known respectively as the Self-Administered Early Grade Reading Assessment (SA-EGRA) and the Self-Administered Early Grade Mathematics Assessment (SA-EGMA) instruments, in Kiswahili, with the support and at the direction of Imagine Worldwide. The assessments are deemed “self-administered,” because children complete the assessments independently in response to instructions and stimuli embedded in a tablet-based software. Adults typically supervise the organization of the assessment and orient the children to the tablet-based assessment environment before they start.

The effort to develop these Kiswahili instruments took place in 2025, beginning with an adaptation workshop with language and curriculum experts in Tanzania in May, where items for the SA-EGRA and SA-EGMA were adapted and revised based on the needed specifications established in previous SA-EGRA and SA-EGMA development efforts. The instruments then underwent user testing with 20 Grade 2 students. Minor changes were made to the user interface and instruments in response to the feedback from user testing. The revised instruments were field tested in Grades 2 and 4 at 20 schools across five regions in mainland Tanzania with 374 students taking the SA-EGRA and 371 students taking the SA-EGMA. Item response theory (IRT) and factor analysis from this field test led to further minor alterations to the instruments, including removal of several lower performing items. The next versions of the instruments were pilot tested from September 22, 2025, through October 2, 2025, with 600 students across Grades 2, 3, and 4 in 15 schools in five regions of Tanzania to assess internal consistency, reliability, and concurrent validity with the traditional EGRA and EGMA instruments. This report presents the results of this pilot testing phase and recommendations for the final instruments to be made publicly available in early 2026.

The findings are encouraging, and mirror past efforts of building self-administered literacy and numeracy assessment tools (for Chichewa in Malawi and English in West Africa), showing high internal consistency across tasks in the instruments and generally acceptable internal consistency within tasks. Both SA-EGRA and SA-EGMA performed well on test-retest reliability, showing students mostly scored consistently across timepoints.

The self-assessment instruments differ from the traditional EGRA and EGMA in that they do not assess oral reading fluency (but focus on accuracy of understanding in the case of EGRA), nor do they measure numeric automaticity as is the case for the traditional EGMA. Thus, some of the constructs being measured by the self-assessment versus the traditional EGRA and EGMA are slightly different. As a result, the correlation between the instruments is reduced, but generally acceptable.

Lastly, analysis of this pilot test has found that there are many literacy items that are working well within their respective tasks and should be reduced. The average time for a student to take an assessment, even after being introduced to the format, is still too long and could create fatigue. This report provides additional analysis to inform such future reductions.

We conclude the report with specific recommendations for future, albeit small changes to the instruments to create the final versions. We also propose continued innovation and research to inform potential refinement of administration protocols, tasks, and items of the SA-EGRA and SA-EGMA instruments.

1 Introduction and Background

This report presents findings from the pilot test of the self-administered EGRA (SA-EGRA) and EGMA (SA-EGMA) in Tanzania in September/October 2025. The purpose of the pilot test, one of three data collections that were part of the instrument development process, was to evaluate the reliability and validity of the SA-EGRA/SA-EGMA currently being developed for Kiswahili. An instrument's reliability is its ability to measure the desired construct with consistency. The instrument should measure consistently both across time and across items. An instrument's validity is the extent to which it measures the construct of interest. For the purposes of this evaluation, the constructs we were interested in measuring were early literacy and mathematics skills.

This report details the performance of the SA-EGRA and SA-EGMA instruments being piloted to determine their fit for purpose and psychometric properties, for the later purpose of using them to reliably and validly evaluate student learning outcomes in foundational literacy and numeracy in Kiswahili in the Tanzanian context.

This report summarizes how the Kiswahili SA-EGRA and SA-EGMA were developed; the data obtained from our pilot test; and the instruments' psychometric properties. The conclusion presents recommendations for the use of the instruments as well as avenues for further development and refinement for various use cases.

2 Assessment Framework

Understanding the psychometric properties of the self-assessment instruments enables us to make changes to their constituent tasks and items that would improve overall reliability and validity. We designed the study accordingly. The next section provides a high-level overview of the key measures of interest we sought to understand.

2.1 Reliability

To determine whether a tool can produce consistent results *over time,* students need to be assessed at least twice using the same tool. This is termed a "test-retest" method. The two assessments need to be conducted relatively close in time (e.g., a few days apart) so the results are not influenced by changes in the student's actual learning. However, they must not happen too closely in time (e.g., the same day) lest the students recall their responses to the first assessment and rely upon that familiarity with the items during the second assessment. We retested students one week following the first assessment.

Two statistical tests are used for test-retest reliability. The first of these is a simple correlation using Pearson's *r* (the Pearson product-moment correlation), a measure of the generalized linear association between two sets of data. An association of 0.5 or higher is considered strong, and an association between 0.3 and 0.5 is considered acceptable. The second approach is the Bland-Altman[1] analysis, which assesses the level of agreement between pairs of repeated measures across the spectrum of the student ability levels. In this approach, if the level of agreement between measures is consistently less than two standard deviations apart, it is considered strong. Reliability can also be measured across tasks or items (rather than over time). If some items within a task appear to be assessing different constructs, removing or replacing them may yield a task that more consistently (reliably) assesses a single construct. Factor analysis measures how closely items within a task (or tasks within an assessment) are related to each other. Generally, the factor loadings are considered acceptable at a level of 0.3 or higher.[2] We apply factor analysis both within tasks (at the item level) and across tasks (to assess reliability of the overall instruments).[3]

The items within each task can also be assessed using Item Response Theory (IRT). In classical test theory, the students being assessed are the unit of analysis. In IRT, the test itself is the unit of analysis. If the items within a task measure the same construct, they can be compared in terms of their difficulty, their ability to discriminate between students of different abilities, and their bi-serial correlation (the correlation between students' scores on the item and total scores on the task).

2.2 Validity

The validity of a tool is the degree to which it *actually* measures what it has been designed to measure. The SA-EGRA/SA-EGMA are designed to measure early grade literacy and numeracy skills, respectively; the scores students achieve on each should therefore be associated with the scores they achieve on the traditional assessor-administered EGRA/EGMA.

Relative to the traditional EGRA/EGMA, the SA-EGRA/SA-EGMA introduce differences in the medium of assessment (tablet-based stimuli vs. paper stimuli) and the protocol (self-administered vs. assessor-administered) while presenting some limitations in task design (e.g., the absence of fluency measures). By adopting an approach called concurrent validity—having the same student complete both the traditional and self-administered versions of the assessments—we can explore the extent to which the SA-EGRA/SA-EGMA can measure our constructs of interest despite these differences. We measured concurrent validity between the two assessments using Pearson's r and looking for an association of 0.5 or higher for a strong association or between 0.3 and 0.5 for an acceptable association between the two assessments.

[1] Bland, Martin J., and Altman, Douglas G., 1986. "Statistical Methods for Assessing Agreement Between Two Methods of Clinical Measurement." The Lancet 327 (8476): 307–10. https://doi.org/10.1016/S0140-6736(86)90837-8
[2] Costello, Anna B, and Osborne, Jason W., 2005. "Best Practices in Exploratory Factor Analysis: Four Recommendations for Getting the Most from Your Analysis." Exploratory Factor Analysis 10 (7): 9.
[3] Because Pearson's correlation and factor analysis are both correlation-based, they have limitations when applied to discrete and binary data rather than continuous measures. For more please see Kolenikov, Stanislav, and Gustavo Angeles. 2009. "Socioeconomic Status Measurement With Discrete Proxy Variables: Is Principal Component Analysis a Reliable Answer?" Review of Income and Wealth 55 (1): 128–65. https://doi.org/10.1111/j.1475-4991.2008.00309.x

3 Instrument Development Process

3.1 App Development

This report will not revisit in full the item specifications detailed in the earlier document entitled, *Draft Assessment Specifications and Prototype (APK) Ready for Field Testing: Summary of the Task and Item Development for the Pilot Self-Administered Early Grade Reading Assessment (SA-EGRA) and Self-Administered Early Grade Mathematics Assessment (SA-EGMA)* (RTI International, 2022).[4] An adaptation workshop was held in Dar es Salaam, Tanzania from May 12 to May 16, 2025, facilitated by RTI and attended by Tanzanian language and education experts from the University of Dar es Salaam, Tanzania Institute of Education (TIE), National Examinations Council of Tanzania (NECTA), DataVision International, and Imagine Worldwide (TZ team). The experts were gathered with the following workshop objectives:

- Review draft instructions/translations for SA-EGRA and SA-EGMA tasks.
- Develop context, language, and grade appropriate items for tasks.
- Develop audio recordings for instructions and test items.

The adaptation workshop was successful and resulted in tasks and items according to the SA-EGRA and SA-EGMA specifications. These items and accompanying instructions were then recorded by Kiswahili speakers and imported into the Tangerine tool for administration. The sections below provide a high-level description of the tasks and skills assessed by the SA-EGRA and SA-EGMA as well as the overall process by which the instruments were refined.

3.2 SA-EGRA Tasks and Literacy Skills Assessed

Letter Sound Recognition

The *Letter Sound Recognition* task assesses the student's knowledge of letter sound–symbol correspondence at the phoneme level. Unlike on the traditional EGRA, in which the student reads letter sounds aloud from a grid of 100 for up to one minute, on the SA-EGRA this task contains 12 items in a receptive, multiple-choice format. For each item, the student is presented with five written letters on the screen and hears the sound associated with one of them. The student then taps the letter that corresponds to the sound that they hear. The student may tap a button to hear the sound again as many times as they want before answering. Distractor letters are chosen based on either their phonological or visual similarity to the target letter. Between the target letters and the distractors, the task incorporates a broad range of Kiswahili phonemes. On the SA-EGRA, this task is untimed and therefore measures only accuracy of recognition, not fluency.

Syllable Recognition

The *Syllable Recognition* task is structured similarly to the Letter Sound Recognition task; the student is presented successively with 11 items, each featuring five syllables on the screen,

[4] Interested readers may request access to this report.

hears one of them, and taps the syllable that they hear. For the SA-EGRA, this task is untimed and measures accuracy of recognition.

Spelling

The *Spelling* task further assesses the student's ability to apply their knowledge of letter sound correspondences and common spelling patterns to encode words. The task format is a dictation in which the student is asked to transcribe 12 words given orally. For the initial prompt, the student hears the word in isolation, hears it used again in the context of a sentence, then hears it in isolation twice more. They then spell (types) out the word using a virtual alphabet strip. The student can tap a button to repeat the word as many times as desired. The student is given partial credit for partially correct answers. The words on the *Spelling* task were carefully selected to assess a broad range of common phonemes and syllable structures in Kiswahili.

Reading Comprehension (Silent and Oral Reading Comprehension)

The *Short Story Reading Comprehension* task assesses the student's ability to understand a short text. In the SA-EGRA, the student reads a short story (approximately 60 words long) written at the Grade 2 reading level. The student is instructed to read the story out loud to themself. In these respects, the text and the task are like those commonly given to assess oral reading fluency (ORF) on the traditional EGRA, though ORF is not measured on this version of the SA-EGRA. For the pilot test in Tanzania, the student is then presented with six multiple-choice questions about the story, one at a time. Each question has four answer options. The questions and answer options (though not the reading passage itself) are presented both in written and oral form. Because the passage is short, the student must answer from memory and lookbacks are not allowed. To approximate Grade 2 text readability, the texts from the Tanzania's Grade 2 pupil book were analyzed for average word and sentence length, two features associated with text difficulty. The passage on the SA-EGRA was written to similar specifications.

The *Silent Reading Comprehension* task assesses the student's ability to understand a longer text, read silently. The student reads a slightly longer story (approximately 140 words long) written at the Grade 4 reading level. For the pilot test, the student is instructed to read silently to themself and then answer eight multiple-choice questions, one at a time. Again, each question has four answer options. To mitigate the limitations of short-term memory, the student is allowed to look back to the passage at will while answering the questions, though they cannot return to previously answered questions. The questions and answer options (though not the reading passage itself) are also presented both in written and oral form. The same method for approximating Grade 4 text readability was applied using the texts from the Tanzania's Grade 4 pupil book.

Language Proficiency (Vocabulary and Syntax)

Finally, the *Vocabulary* and *Syntax* tasks assess two other important aspects of the student's proficiency in Kiswahili. Assessing language proficiency alongside reading is one way to surface whether any reading comprehension difficulties are due to low language proficiency, a distinction that is especially important in contexts where students are learning to read in languages that they do not speak at home.

The *Vocabulary* task captures data about the breadth of the student's receptive knowledge of Kiswahili words. For the pilot test, the task presents the student with 20 multiple choice items, each with one real word and three pseudoword answer options. The student is prompted to "Tap the word that you know the best." The real word is the only option that the student could know. If the student does not know the target word, they will resort to guessing. All pseudowords conform to Kiswahili phonology and orthography and therefore look like possible words. All words are presented in both oral and written form, and the student can tap a button to hear the word again as many times as desired. The oral stimulus is used to mitigate the possible effects of low reading ability; the written stimulus helps the student hold all four options better in their short-term memory while deciding among them. The target words were "academic" or "text" words that are encountered primarily in text versus in everyday life. Word lists were compiled from Tanzania's Grades 1-4 pupil books and analyzed for frequency. During the adaptation workshop, the team of Tanzanian education experts then examined the word lists and chose an initial list of 24 words, eight first encountered in Grade 1 and 2 texts, eight first encountered in Grade 3 texts, and eight first encountered in Grade 4 texts. The team also developed 72 pseudowords with a similar distribution of word lengths to serve as distractors for the 24 target words (three distractors for each real word). The 24 original items were narrowed down to 20 based on their performance on the field test.

The *Syntax* task assesses the student's knowledge of how words are put together to make meaning in Kiswahili. In the pilot test, the task presents the student with eight items with four short sentences each. All sentences use basically the same vocabulary but differ primarily in syntax (word order) and morphology (word form). One sentence makes sense (e.g., The house has a door.) and the others do not (e.g., The door has a house.). The student is instructed to "tap the sentence that makes sense." The sentences are presented both orally and in written form for the same reasons as mentioned above, and the student can tap a button to hear them again as many times as desired. The sentences contain only common words to minimize the potential of a student's limited vocabulary being a constraint. The items focus on a variety of common Kiswahili syntactic and morphological structures such as adjective-verb agreement with the noun class, sentence word order, verb tense, locative word order, or relative clauses. A low score on the *Syntax* task indicates that one's low Kiswahili language proficiency is likely a hindrance to reading comprehension rather than (just) poor reading skills.

3.3 SA-EGMA Tasks and Mathematical Skills Assessed

As with the SA-EGRA, the SA-EGMA does not currently assess fluency (i.e., automaticity). However, the SA-EGMA assesses the same mathematical skills as the traditional EGMA through the following tasks: *Number Identification, Number Discrimination, Missing Number Identification, Addition, Subtraction,* and *Word Problems*. Most of these tasks are identical to the traditional EGMA, with the only difference being that responses require the child to type their answers instead of saying them aloud to an assessor.

Number Identification

The *Number Identification* (a.k.a. “number ID”) task evaluates students’ ability to connect the spoken name for a number (e.g., “three”) with its symbolic representation (e.g., “3”). Unlike the traditional EGMA, the SA-EGMA sounds out the name of a number (e.g., “three”) and asks students to enter the corresponding symbolic representation (e.g., “3”). This task has 12 items of increasing difficulty, and a keypad with numbers 0-9 for the student to use to enter their answer. When the student has typed in their answer, they press the arrow button to move on to the next item. If the student answers four in a row incorrectly, the task ends and they are presented with the next task. This is the same autostop rule as in the traditional EGMA.

Number Discrimination

The *Number Discrimination* (a.k.a. *Number Comparison*) task evaluates students’ ability to discern between quantities, and to identify the largest number (e.g., 58 is larger than 49 and 32). This task also assesses students’ place-value skills by presenting pairs in which the larger number has a smaller ones or tens digit than the smaller number (e.g., 534 vs. 287 vs. 199). Each item has three options to choose among, and there are 10 items in the task. If the student answers four in a row incorrectly, the task ends and they are presented with the next task.

Missing Number

The *Missing Number* task evaluates students’ ability to identify a missing element in a sequence of numbers (e.g., 14, 15, __, 17). The 10 test items are identical to the traditional EGMA. Instead of asking students to say the name of the missing number aloud, as in the traditional EGMA, the SA-EGMA asks students to enter the missing number using the number keypad of 0-9. The format enables visual verification not afforded by the traditional format. That is, students can see the complete pattern in the self-assessment version whereas they only say the number aloud in the traditional version. If the student answers four in a row incorrectly, the task ends and they are presented with the next task.

Addition and Subtraction

The *Addition* and *Subtraction* tasks assess students’ addition and subtraction skills using single-digit (Level 1) and double-digit (Level 2) numbers. *Addition Level 2* and *Subtraction Level 2* are only administered to students who give at least one correct response to the corresponding Level 1 assessment. The Level 1 tasks contain eight items each and Level 2 tasks contain five items. Initial field and pilot testing had 13 items each for *Addition* and *Subtraction Level 1*. The tasks were reduced as a result of initial pilot analysis to keep the most relevant items to appropriately measure inherent subskills and align length with that of other SA-EGRA and SA-EGMA. All analysis included in this report reflects the reduced tasks. For the analysis of the full 13 items in these tasks, and which items were chosen for removal, refer to ***Annex A***. All *Addition* and *Subtraction* tests are untimed and thus do not assess fluency. The students may use pencil and paper for the Level 2 tests. If the student answers four in a row incorrectly, the task ends and they are presented with the next task. If the student scores zero on either *Addition Level 1* or

Subtraction Level 1, they do not receive the corresponding Level 2 task, as with the traditional EGMA.

Word Problems

The *Word Problems* task evaluates students' ability to understand an arithmetic problem described in narrative form (e.g., "Three students are on a bus. Two more students join them. How many students are on the bus now?") in a way that allows them to operate on and solve the problem (e.g., 3 + 2 = ___, counting on from 3, etc.). Students may use pencil and paper, or manipulatives if available, as they work on each problem, though only the final answer as entered on the tablet is recorded. The six word problems are identical to those on the traditional EGMA. While the tablet cannot use verbal and non-verbal cues to check students' understanding as administrators of the traditional EGMA are trained to do, students may repeat the problem narrative as many times as they wish. This task contains six items, If the student answers four in a row incorrectly, the task ends and they are presented with the next task.

3.4 Stages of Assessment Development and Refinement

The development and refinement of the instruments proceeded in three main stages following the adaptation workshop.

1. **User testing:** this stage occurred during initial and subsequent renderings of the instruments. The instruments were tested with small user groups, with feedback collected both via direct observation and explicit questions.
2. **Field testing:** this stage assessed the initial performance of the instruments, with the objective to refine them and address issues prior to piloting. For the SA-EGRA and SA-EGMA, the field test included analysis of multiple-choice items, assessment duration, protocols, and overall construct validity.
3. **Pilot testing:** in this stage, the instruments were fully assessed for their psychometric properties, with a particular focus on reliability and validity. The analysis from the data collected from this assessment stage form the basis of the main findings detailed in this report.

User Testing

The user test of the instruments took place July 21–24 in two primary schools, Kisarawe (rural - Kazimzumbwi) and Kinondoni (urban- Msasani B). Imagine partner DataVision in Tanzania conducted systematic user testing with 20 Grade 2 students to help localize and refine the user interface, workflows, and software interactions to the local context. Following that, we made small revisions to these elements and readied the assessments for field testing.

Field Testing

The field test of the instruments took place August 19–23. DataVision assessed 785 students in Grades 2 and 4 in 20 schools in Ruvuma (Mbinga TC), Tabora (Nzega DC and Tabora DC), Shinyanga (Kahama TC and Msalala DC), Manyara (Hanang DC), and Morogoro (Gairo DC and Kilosa DC).

This report will not revisit in full the analysis of the field test, as that is already detailed in the earlier document entitled *Kiswahili Field Test Report* prepared by RTI and submitted to Imagine Worldwide. In summary, however, data analysis of field test results suggested that for the reading assessment, the task percent scores performed well on the confirmatory factor analysis, all loading highly on a single factor, and all had good Item-Test and Item-Rest Correlation scores within the Cronbach's Alpha analysis. The overall Cronbach's Alpha was 0.87, which is very good. *Letter Sounds* performed slightly lower than the other tasks, which may have more to do with the task requiring a lower level of literacy ability than others. Several items, however, were marked for removal or change, following more detailed analysis. As a result, before the pilot test, the team removed one *Syllable* item, one *Short Reading Comprehension* item, four *Long Reading Comprehension* items, four *Vocabulary* items, and one *Syntax* item. Changes were also made to Short Reading Comprehension Item 2, the introduction of the *Syntax* task, and to *Syntax* Item 6 to enhance clarity in wording.

For the mathematics assessment, the numeracy task percent scores also performed well on confirmatory factor analysis, all loading highly on a single factor. All tasks also had reasonable Item-Test and Item-Rest Correlation scores within the Cronbach's Alpha analysis. The overall Cronbach's Alpha was 0.90, which is excellent. No changes were made to the math assessment in advance of pilot testing.

Pilot Testing

The pilot test of the instruments was conducted from September 22, 2025, through October 2, 2025, with data again collected by DataVision. Pilot testing included two components for validation. The first component was for concurrent validity, and this involved the same student being assessed with both a traditional EGRA or EGMA and the self-administered counterpart on the same day. The second component was for test-retest reliability and involved the same students being retested with the SA-EGRA or SA-EGMA after a period of six days. A total of 600 students took part in the pilot test, 300 each for the SA-EGRA and SA-EGMA, across Grades 2, 3, and 4 in 15 schools across Ruvuma (Songea DC), Tabora (Nzega DC and TC), Shinyanga (Shinyanga TC & DC), Manyara (Babati TC & DC), and Morogoro (Ifakara TC and Ulanga DC).

To best assess the performance of the instrument across all ability ranges, we sought a uniform distribution of student abilities in the reading assessment and the fewest zero scores possible. The concurrent validity approach allowed us to monitor ORF scores on the traditional EGRA and adjust the student selection process to obtain the desired range of student abilities. Because of this, most of the sample were students from Grades 3 and 4, although the instrument was created to align with the curricula of Tanzanian Grades 2 and 4.

Both RTI and Imagine Worldwide (with DataVision) followed appropriate Internal Review Board (IRB) procedures at RTI and in Tanzania, respectively. In Tanzania, the team obtained a research permit from the Commission for Science and Technology (COSTECH) and the President's Office – Regional Administration and Local Governments (PORALG).

4 Pilot Test Findings

4.1 Student Literacy Outcomes

The outcomes from the literacy assessment should be viewed within the context of the sampling approach used. The students were purposefully selected to create a uniform distribution of student performance in oral reading fluency and to avoid zero scores. Because of this, the average percent scores shown in **Exhibit 1** for the SA-EGRA are not necessarily representative of the general population of students in the early grades.

Exhibit 1. SA-EGRA Pilot Average Learning Outcomes, by Task

SA-EGRA Task	Average Percent Score
Syntax	64.4%
Letter Sounds	83.5%
Vocabulary	68.4%
Spelling	68.9%
Silent Reading Comprehension	66.9%
Short Story Reading Comprehension	67.0%
Syllables	80.2%

The general pattern of performance on these tasks can still be meaningfully compared to a traditional EGRA, as higher scores generally map to lower-order skills and vice versa. Students are scoring highly in *Letter Sounds* (83.5%) and *Syllables* (80.2%) but are struggling more with the higher-order skills involving reading comprehension. *Silent Reading Comprehension* had an average score of 66.9% and Short Story Comprehension had an average score of 67.0%. Interestingly in this assessment, students struggled more with the *Syntax* task (64.4%) than expected for native Kiswahili speakers. That task is thought to be best aligned with the skills of listening comprehension, which should be a lower-order literacy skill. Further analysis in this report explores why this might be the case.

4.2 Student Mathematics Outcomes

While students who were assessed in literacy were purposefully sampled to ensure a roughly uniform spread across literacy levels, students who were assessed in mathematics were more randomly selected from the visited classrooms across Grades 2, 3, and 4. As with the literacy assessment, the average percent scores presented in **Exhibit 2** follow similar performance patterns seen in traditional EGMA tasks, with scores tending to decrease as difficulty increases. As in previous self-administered instruments, the *Number Identification* task stands as an outlier to the general pattern, with students scoring lower than expected for the difficulty of the task. This is the first task students encounter in the self-administered format,

Exhibit 2. SA-EGMA Pilot Average Learning Outcomes, by Task

SA-EGMA Task	Average Percent Score
Number Identification	67.7%
Number Discrimination	78.2%
Missing Number	51.1%
Addition Level 1	80.2%
Addition Level 2	63.6%
Subtraction Level 1	71.8%
Subtraction Level 2	55.7%
Word Problems	44.8%

and as such, these responses are likely indicative of students learning the input features of the tablet rather than entering incorrect responses. There are also more instances of students entering long strings of multiple digit numbers in this task compared to others. This behavior was observed across all student ability levels, even among those who scored highly in other subtasks. While this is an interesting artifact of the self-administered format, it is heartening to know that average scores on this task were much higher (77.7%) at Timepoint 2 of the pilot test, and the instances of random number sequences were much less. This leads us to believe that students who are more exposed to the assessment will be less likely to repeat these patterns. This behavior at the first instance of assessment will, however, need to be considered in future uses of the instrument for overall scoring purposes.

4.3 Time Taken to Complete Assessment

One of the critical considerations for use of this assessment is the time required for a student to complete the assessment. Aside from logistical implications, test fatigue can affect performance. Average duration of student assessment at each timepoint is presented in **Exhibit 3**.

Exhibit 3. Mean Assessment Durations for SA-EGRA and SA-EGMA, by Timepoint

Assessment	Mean (mins.)	Standard Deviation
SA-EGRA (t_1)	69.1	±21.6 mins
SA-EGRA (t_2)	51.8	±18.2 mins
SA-EGMA (t_1)	58.6	±20 mins
SA-EGMA (t_2)	41.8	±15.4 mins

As shown, the average duration of the assessments is reduced substantially between timepoints, which may point to a learning curve in using the assessments. Although the time taken is reduced at Timepoint 2, the average duration of the assessment is still very long. We recommend shortening the assessments wherever possible to prevent test fatigue. Consequently, we have implemented autostop rules for the SA-EGRA to help students who cannot read to move through the assessment more quickly following this pilot test analysis. We further recommend removing items to shorten tasks, which are described in more detail in later sections of this report. We also suggest additional training for supervising adults to advise students to move on if they are struggling.

4.4 Internal Consistency

Internal Consistency of the SA-EGRA Tasks

Here we present a summary of the instrument's internal consistency; the detailed analyses are provided in **Annex D**. We assess the overall internal consistency for the SA-EGRA using Confirmatory Factor Analysis of the task percent scores. This provides an opportunity to assess if the tasks were measuring the same latent construct. Our analysis summarized in **Exhibit 4** found strong factor loadings onto a single construct ranging from 0.4916 to 0.8465. Given that acceptable factor loadings should be 0.3 or more, the internal consistency at the summary level for the SA-EGRA is excellent. While the factor loadings for most tasks are very high, the factor loadings for *Letter Sounds* is noticeably lower. As this task is performing similarly to how it has performed in other assessments, it is still considered a good measure of literacy, but not quite in line with the other tasks. The current assumption is that this is because the task involves a lower-order skill for students while the selected sample of students for this pilot were overall higher-performing in literacy. We can see by the scores for the task that this is most likely the case; each item in the task shows 80% or more students answering it correctly. This task will more than likely perform as expected with younger students, or in a context with a wider range of literacy abilities.

Exhibit 4. Factor Analysis Loadings or SA-EGRA Task Scores

SA-EGRA Task	Factor 1 Loadings
Letter Sounds	0.4916
Syllables	0.6755
Short Story Reading Comprehension	0.7597
Silent Reading Comprehension	0.7957
Vocabulary	0.8465
Syntax	0.8186
Spelling	0.8067

Each of the SA-EGRA tasks was also assessed for internal consistency at the item level using both factor analysis and Item Response Theory (IRT). We present the analysis for the *Syntax* task in **Exhibit 5** and discuss it as an example. For the sake of brevity, the comparable tables for the other tasks are included in **Annex C**. The discussion here will be limited to key points in summary form.

Exhibit 5. Item Factor Analysis and IRT for the Syntax Task

Item Number	Factor Analysis	Item Response Theory		
		Discrimination	Difficulty	Bi-serial Correlation
1	0.7273	0.9	0.63	0.79
2	0.5949	0.61	0.81	0.66
3	0.5994	0.79	0.6	0.67
4	0.6743	0.83	0.72	0.7
5	0.6600	0.85	0.61	0.72
6	0.5819	0.81	0.65	0.63
7	0.6210	0.63	0.82	0.66
8	0.5942	0.71	0.7	0.67

For IRT to work well, the items being analyzed should explain the same construct. To ensure this is the case, we first perform Factor Analysis, to ensure that all items have a factor loading over 0.3. This is the case here, which suggests that all items are working well together on the same latent construct. This is also the case across all the SA-EGRA subtask, as all items tested had factor loadings over 0.3, indicating that all subtasks measure the same construct, and IRT analysis is appropriate.

The IRT item-level analysis for *Syntax* starts with Discrimination. This reports the difference between the proportions of high and low scorers answering an item correctly. A Discrimination score of over 0.2 helps an item to contribute toward reliable measurement across a range of student ability. The discrimination scores for the *Syntax* items are all acceptable, ranging from 0.61 (Item 2) to 0.9 (Item 1). Item Difficulty is the proportion of students who correctly answered that item. It ranges from 0 (no student answered correctly) to 1 (all students answered correctly). For an assessment designed specifically to measure varying student skills, difficulty scores should ideally range from 0.2 to 0.8. Most of the *Syntax* items fall on the higher end of this range, with two of the items exhibiting Difficulty scores just over 0.8. *Syntax* was the second highest-scoring task behind *Letter Sounds,* so these Difficulty scores indicate this is a generally easier task than most others. The bi-serial correlation is the Pearson correlation between responses to a particular item and scores on the overall task. It ranges from -1 to 1, and strong positive correlations are desirable. The bi-serial correlations for the *Syntax* items are acceptable, ranging from 0.63 to 0.79. Altogether, these analyses point to a suitably strong task that can be used to assess lower-level literacy skills in students.

Internal consistency for *Letter Sounds* was acceptable overall (***Exhibit C2 of Annex C***). All factor loadings were in the acceptable range, with each item having appropriate levels of Discrimination and Difficulty, although there was not a wide range across these measures, and they tended to be on the lower end for Discrimination and the higher end for Difficulty. This lack of variability in Discrimination and Difficulty is most likely due to the task targeting lower-order literacy skills, and our sample population needing to consist of mostly Grade 3 and 4 students.

The internal consistency for the *Vocabulary* task is shown in ***Exhibit C3*** of ***Annex C***. The Discrimination and Difficulty scores are acceptable, with only item 15 having a Difficulty score right at 0.8. We do see some more difficult items in this task (Items 4, 9, 11, and 12), so this is an optimal spread for a lower-order skills task. All factor loadings fall into the acceptable range for good internal consistency. This task was created with a large bank of items in the understanding that they would be pared down based on results of the field test and pilot test. There are still 20 items remaining that are performing well. Previous instruments in other contexts and languages had only 10–14 items remaining at this stage. For future administration of the SA-EGRA, we recommend reviewing the items with counterparts familiar with the context and selecting items for removal that are performing similarly in terms of Discrimination and Difficulty. This will help to prevent test fatigue among students. Specific analysis to inform such choices is in **Annex B.**

The *Syllables* task (**Exhibit C5** of **Annex C**) also shows good internal consistency, with factor loadings all falling within the acceptable range. Discrimination scores all fall above 0.2, with Item 5 being the lowest at 0.28. Item 5 also has a Difficulty score of 0.92 (meaning 92% of students answered it correctly), so it stands out as a much easier item for this task and sample of students. Overall, the Difficulty scores are on the higher end, with 7 of the 11 tasks (Items 4, 5, 6, 8, 9, 10, and 11) having scores of 0.8 or higher. This indicates that this task is generally quite easy for this sample of students but may be suited for a wider population of Grade 1–4 Tanzanian students.

For the *Short Story Reading Comprehension* task (**Exhibit C6** of **Annex C),** all factor loadings are above 0.3, and all Discrimination and Difficulty scores fall within the acceptable ranges.

There is not a wide range of Difficulty or Discrimination scores, but with only six items on a higher-order skill task, this is reasonable.

The internal consistency for the *Silent Reading Comprehension* (**Exhibit C7** of **Annex C**) task was acceptable overall, with all factor loadings over 0.3. Discrimination scores were all above 0.2, with Item 4 showing a quite high score (0.9). The Difficulty scores ranged from 0.52 (Item 4) to 0.87 (Item 2), which indicates a good spread of difficulty across items in this task.

The *Spelling* task has different item characteristics than the others in the SA-EGRA instrument. Other tasks present the student with a binary or multiple choice and score it as correct or incorrect, while the *Spelling* task requires students to actively produce text and the scoring mechanism awards partial credit for incorrect responses. Three-parameter IRT analysis—which primarily deals with binary items that are fully correct or fully incorrect—is therefore not suitable for this task. The item factor analysis (***Exhibit C8*** of ***Annex C***), however, is excellent, with factor loads between 0.6648 and 0.8957, easily surpassing the 0.3 threshold. Many different factors contribute to the strength of this task. From a psychometric perspective, this task (unlike the others) provides minimal opportunity for correct guessing; can discriminate between different skill levels by awarding partial credit; and can yield an aggregate score between 0 and 91 (a wide continuous measure). Consequently, the *Spelling* task is excellent for capturing the variability of students' skill levels. This is another task where more items were piloted than intended for final use, but they are all still performing well. There are 12 items in this task, while previous assessments only had 8 and 10. To reduce assessment burden and fatigue, we recommend removal of two to four *Spelling* items that are similar to others in the task for future administrations. Specific analysis to support such effort, ideally to be done by Kiswahili language and Tanzania Grades 1-4 curriculum experts, is in **Annex B.**

Overall, the internal consistency within the SA-EGRA instrument is strong and the analysis points to a set of cohesive tasks that are measuring the same general construct of literacy. This, combined with the internal consistency of the full SA-EGRA assessment, points to a strong instrument for assessing literacy among Kiswahili-speaking students in Grades 1-4 in Tanzania.

Internal Consistency of the SA-EGMA Tasks

As with the SA-EGRA instrument, we assessed the overall internal consistency of the SA-EGMA using factor analysis of the task percent scores. The analysis, summarized in **Exhibit 6**, found moderately strong factor loadings onto a single construct ranging from 0.6268 (*Number Identification*) to 0.7926 (*Addition Level 2*). Given that acceptable factor scores should be 0.3 or more, the internal consistency of the tasks for the SA-EGMA is good and the tasks are all performing well together to measure numeracy.

Exhibit 5. Factor Analysis Loadings for SA-EGMA Task Scores

Task Percent Score	Factor 1 Loadings
Number Identification	0.6268
Number Discrimination	0.7203
Missing Number	0.7732
Addition Level 1	0.6738
Addition Level 2	0.7926
Subtraction Level 2	0.6403
Subtraction Level 2	0.7269
Word Problems	0.6958

IRT techniques are a poor fit for free-response item types. Except for *Number Discrimination*, all SA-EGMA tasks were presented in a free-response format. As a result, we did not perform IRT analyses of the SA-EGMA. Item-level internal consistency analyses for the SA-EGMA were limited to factor analysis, shown in detail in **Annex C**.

Exhibit 7 presents the item-level factor analysis for the *Missing Number* task as an example. While this task was on the higher end of factor loadings at the task level, its item-level characteristics are generally similar to those of the other tasks. Six of the 10 items surpass the target threshold (a factor loading of 0.3 on the first factor). While four of the items show factor loadings under 0.3, these scores are not out of the ordinary for traditional EGMA assessments as explained below.

Exhibit 6. Item Factor Analysis for the Missing Number Task

Item	Factor Analysis
1	0.1995
2	0.2305
3	0.3351
4	0.1405
5	0.6726
6	0.1454
7	0.5891
8	0.5259
9	0.4096
10	0.5457

Item-level factor analysis of the other SA-EGMA tasks reveals similar patterns. Roughly one-quarter to one-half of the items in each task fall below the threshold of 0.3 for first factor loadings. Many are substantially lower than the threshold (ranging from 0.2004 to as low as -0.2459). While the factor loadings for individual items within a task are not very high, this is common in traditional EGMA assessments as well and thus to be expected. The reason is that, within tasks, items are designed to measure easier to harder tasks. This progression within tasks means that those items are not designed to measure exactly the same skill, but rather a progression of a particular skill. For example, within the *Number Discrimination* task, each item is increasingly more difficult, moving from simple discrimination of single-digit numbers to double-digit numbers, and then triple-digit numbers, with different specifications for the types of numbers used in the task according to developmental progressions. Given this, lower factor loadings within a task are not cause for concern but can still provide information on where improvements may be made. As with SA-EGRA items that have low factor loadings, further analysis and investigation is warranted before deciding that items with low factor loadings should be revised. In general, the SA-EGMA tasks are acceptable and fit for the purpose of measuring numeracy skills in Grades 1-4 Tanzanian students.

4.5 Test-Retest Reliability

Test-Retest Reliability of the SA-EGRA Tasks

We assessed test-retest reliability using Pearson’s correlation to report the generalized relationship between student scores assessed at the two timepoints. Pearson’s correlation is generally used when reporting linear associations of continuous variables. When applied to variables with discrete outcomes and a relatively limited number of items, as is the case with the SA-EGRA tasks, lower Pearson’s correlations are to be expected. An acceptable range for the Pearson’s coefficient would be between ±0.3 and ±1, with 0.3 to 0.49 indicating a moderately strong relationship and 0.5 to 1 indicating a high correlation. **Exhibit 8** reports the correlations for student scores on the same task at the two timepoints of the pilot test; the graphs depicting individual students' scores are presented in **Annex D, Exhibit D2**.

Exhibit 7. Pearson's Correlation for the SA-EGRA Test-Retest

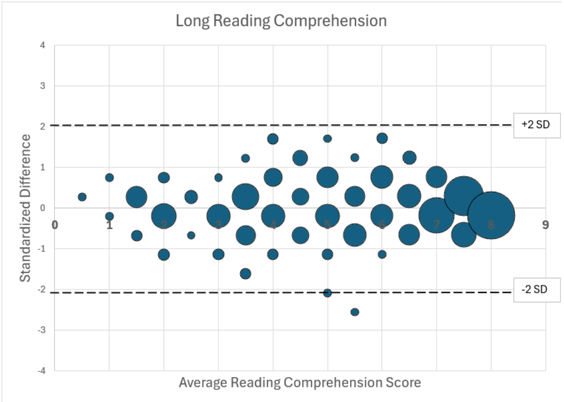
SA-EGRA Task Percent Score	Correlation
Letter Sounds	0.7519
Short Reading Comprehension	0.7297
Syllables	0.7314
Syntax	0.8529
Long Reading Comprehension	0.8229
Vocabulary	0.8672
Spelling	0.9234

The *Spelling* task demonstrates the strongest positive correlation at the two timepoints at 0.9234. As mentioned above, Pearson's correlation analysis is best suited to continuous variables, and the task's continuous scoring and wide range help counteract some of the limitations that this analysis encounters with the other tasks. *Spelling* also demonstrated excellent internal consistency, which supports the conclusion that the high Pearson's coefficient is measuring the correlation of this task appropriately, and that the lower coefficients of other tasks may be more influenced by the discrete scoring of the items and limitations of the analysis (see **Exhibit 4** and **Annex C, Exhibit C8**). The other tasks demonstrate high correlations between 0.7297 to 0.8672.

As mentioned, the limited number of items and discrete scoring of the non-*Spelling* tasks poses a challenge for interpretation of Pearson's correlation. We therefore additionally include Bland-Altman analyses. Bland-Altman assesses the level of agreement between pairs of repeated measures across the spectrum of the student ability levels, rather than the relationship of two sets of results as with Pearson's correlation.

To illustrate this principle, **Exhibit 9** presents the Bland-Altman graph for students' scores on the *Long Story Reading Comprehension* task. The students' average score across the two timepoints is plotted along the horizontal axis. The standardized difference between the students' scores at the two timepoints (score at t_2 – score at t_1) is plotted along the vertical axis.[5] For normally distributed data, we would expect 95% of standardized differences to be within ±2 standard deviations (represented on the graph by the two dotted lines). This is the case for all the SA-EGRA tasks, including *Long Story Reading Comprehension*. The Bland-Altman graphs for the other tasks are included in **Annex D**.

Exhibit 9. Bland-Altman Graph for SA-EGRA Test-Retest, Long Story Reading Comprehension Task



[5] If a student scores higher at t_1, they will be represented by a point below the horizontal zero axis. The score difference is then standardized (i.e., converted to standard deviations).

The Bland-Altman plots and Pearson correlation plots indicate that the SA-EGRA tasks demonstrate very good test-retest reliability (**Annex D**). Notably, only a small percentage of students scored very well at one timepoint of a task and very poorly at the other. (The points representing these students tend to cluster near the center of the horizontal axis and beyond the ±2 bounds. They are present both above and below the 0 line, indicating that t_2 was not uniformly the higher scoring timepoint for these students.) We suspect this phenomenon is likely driven less by the instrument itself and more by how the student approached the assessment at one of the timepoints. As this phenomenon accounts for less than 4% of students at the most, we do not believe it will cause issues with data collection or overall inaccuracies with the assessment but may be an opportunity for future improvements to the instrument. While it may not be possible to fully eliminate the underlying cause, a modification to the administration protocol or instructions may mitigate it.

Test-Retest Reliability of the SA-EGMA Tasks

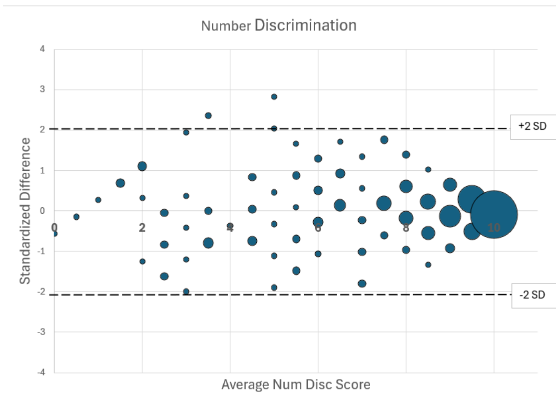
We conducted the same test-retest reliability analyses for SA-EGMA as for SA-EGRA. **Exhibit 10** reports the correlations for student scores on the same task at the two timepoints; the graphs depicting the individual students' scores are presented in **Annex D, Exhibit D5**.

Exhibit 10. Pearson's Correlation for the SA-EGMA Test-Retest

SA-EGRA Task Percent Score	Correlation
Number Identification	0.7301
Number Discrimination	0.8002
Missing Number	0.9240
Addition Level 1	0.8687
Addition Level 2	0.7129
Subtraction Level 1	0.7360
Subtraction Level 2	0.6377
Word Problems	0.7880

None of the SA-EGMA tasks involved either continuous measures or very large numbers of items, but they still showed very high positive correlations. As with the SA-EGRA instrument, caution should be taken in over-interpreting Pearson's correlations; we also complement this analysis with Bland-Altman plots. The plot for *Number Discrimination*, presented in **Exhibit 11**, displays similar patterns as observed with the SA-EGRA tasks. It is also generally illustrative of the other SA-EGMA tasks.

Exhibit 11. Bland-Altman Graph for SA-EGMA Test-Retest, Number Discrimination task



As shown, most students fall within the expected range of ±2 standard deviations, with less than 1% falling outside the range. The larger points, indicating a larger percentage of students, cluster around 0 on the vertical axis, particularly at the higher end of the horizontal axis, indicating that many students scored similarly high at both timepoints. There is a bit of spread of smaller points ranging upward and downward, showing that some students scored high at one timepoint and low at another, but these are all still within the range of tolerance. The Pearson's correlation graph in **Annex D, Exhibit D5** shows a similar spread of scores as well, especially in the middle to lower ranges. This is understandable, as this was the only multiple option task in the SA-EGMA instrument, and students who are just learning number discrimination skills may not always choose the correct option. Aside from this, the *Number Discrimination* task performs very well and can be used as a good metric for assessing number sense in students.

The only SA-EGMA task that is showing any issue is *Number Identification*. The Pearson's correlation graph in **Annex D, Exhibit D5** shows a pattern of students scoring low at the first timepoint, and much higher at the second. This is most likely due to students orienting themselves to the assessment, as this is the first task they encounter. In the data, we do see many responses to these items that are long strings of random numbers, including from students who score well in other tasks. This does occur less at the second timepoint, suggesting that students are experimenting with the number entry functions at the start of the assessment, but after they are familiar with the assessment, this is no longer of interest. This should be taken into consideration when using this assessment, as this task may not always show good results for students who are not familiar with the assessment format.

In summary, the Test-Retest Reliability of the SA-EGRA and SA-EGMA tasks show that all are within the acceptable range, and each task is a reliable measure of the literacy or numeracy construct they are measuring, with very few caveats.

4.6 Construct Validity

Construct Validity SA-EGRA Tasks

Construct validity is typically used to compare a new tool to an existing tool that has been shown to measure the same construct. A high level of association between the new and existing tools suggests that they indeed measure the same construct. However, the administration modality of the SA-EGRA differs substantially from that of the traditional EGRA; comparing mostly multiple-choice tasks (on the SA-EGRA) with oral responses to grids of items (on many traditional EGRA tasks) is not comparing like-for-like. Additionally, there is interest to know if the SA-EGRA is comparable to the traditional EGRA's oral reading fluency (ORF) task, due to the latter's extensive use as measure of program impact and reporting against the United Nations' SDG 4.1.1a.[6] We

sought a method of exploring SA-EGRA's concurrent validity with the traditional EGRA that would likely meet the needs of the SA-EGRA's target user base. We decided to explore whether the SA-EGRA can be used as a proxy measure for ORF by creating a composite SA-EGRA score that combines the task percent scores, weighted according to relative importance derived from expert judgment.

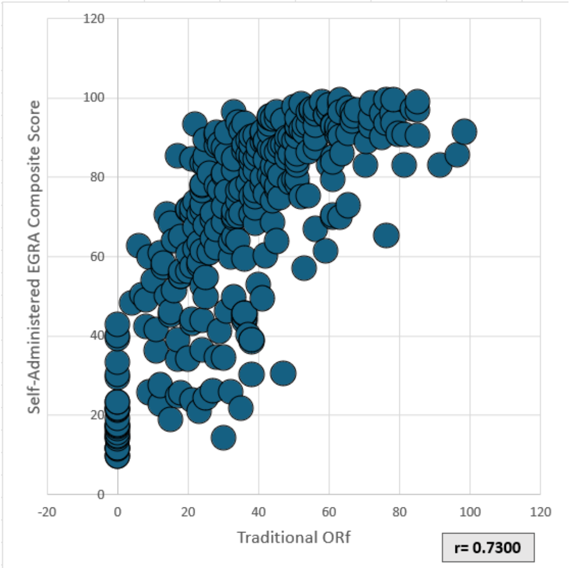
[6] United Nations. 2019. "SDG Indicators." SDG Indicators. Retrieved December 2022. https://unstats.un.org/sdgs/metadata/?Text=&Goal=4&Target=4.1.

Exhibit 12 shows the correlation scatterplot comparing the scores on the SA-EGRA Composite Score and the traditional EGRA's ORF task. The SA-EGRA Composite Score was calculated using:

$$\text{SA\_EGRA\_composite} = 0.4 \times \text{spelling\_total\_score\_pcnt} + 0.15 \times \text{short\_read\_comp\_score\_pcnt} + 0.1 \times \text{long\_read\_comp\_score\_pcnt} + 0.05 \times \text{letter\_sounds\_score\_pcnt} + 0.1 \times \text{vocab\_score\_pcnt} + 0.1 \times \text{syntax\_score\_pcnt} + 0.1 \times \text{syllables\_score\_pcnt}$$

The correlation of r = 0.7300 indicates a fairly strong positive linear association between the two tasks. While this association is strong, it is also important to explore the predictive ability of the model. For example, students who scored a 60 (rounded, 59.5-60.4) Composite Score recorded ORF scores of between 6 and 59 correct words per minute. This suggests that while this statistical linking of SA-EGRA *Composite Score* results with traditional EGRA ORF scores could be used for generalized equivalent findings (such as population estimates), it lacks the precision to provide a 1:1 mapping between SA-EGRA and traditional EGRA for individual students.

Exhibit 12. SA-EGRA Composite Score vs. ORF on the Traditional EGRA



Based on the excellent performance of the *Spelling* task, we also elected to assess how well it is associated with ORF on the traditional EGRA, and those results can be seen in **Annex E**. The statistical linking to this task ended in similar conclusions, that it is good for generalized results of a large group, but not for individual students, although the positive linear association was slightly weaker (r = 0.6361) than for the composite score (r = 07300).

Having performed this analysis now across three contexts, we have determined that the SA-EGRA *Composite Score* is the better option for using a single metric for literacy from this assessment. Although here we have repeated the use of the same composite score, it is possible that there could be better results from a slightly different weighting of the tasks, as many tasks performed differently in this context. We recommend using the SA-EGRA *Composite Score* only as an appropriate measure for literacy and mapping to traditional EGRA scores for population (group) estimates. We also recommend further investigation to be done, along with literacy experts familiar with the Tanzania Grades 1-4 context, to explore adjusting the weights of the composite score to see if it is possible to improve the fit.

Construct Validity SA-EGMA Tasks

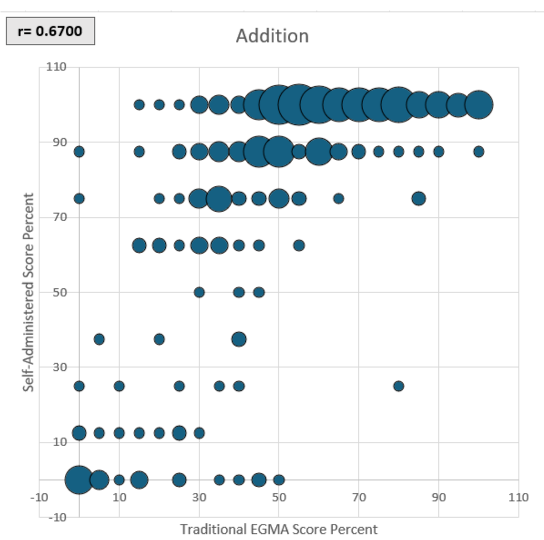
There is a very tight coupling at both the task and item levels between the SA-EGMA instrument and the traditional EGMA. As a result, using Pearson's correlation for generalized assessment of concurrent validity is a more appropriate method than was the case for the SA-EGRA and traditional EGRA. **Exhibit 13** presents the Pearson's correlation for each task. As with the test-retest correlations, an acceptable range for the Pearson's coefficient would be between ±0.3 and ±1, with 0.3 to 0.49 indicating a moderate correlation and 0.5 to 1 indicating a strong relationship.

Exhibit 13. Pearson's Correlation for Generalized Concurrent Validity of the SA-EGMA and Traditional EGMA

SA-EGMA Task Percent Score	Correlation
Number Identification	0.5592
Number Discrimination	0.7296
Missing Number	0.8948
Addition Level 1	0.6700
Addition Level 2	0.5452
Subtraction Level 1	0.6674
Subtraction Level 2	0.5358
Word Problems	0.7296

Much of the SA-EGMA task-level correlations with the traditional EGMA are in the range of 0.55–0.75 range, indicating a fairly strong relationship. To note is the high correlation (0.8948) of the *Missing Number* task, while the *Addition* and *Subtraction* tasks are more weakly correlated. This follows patterns in previous self-administered assessments, where the differences in administration modality have a substantial influence over some math scores. With the traditional EGMA, having an assessor sitting with an individual student, timing them and nudging them onwards through the task, as well as timing the student for fluency, appears to have a notable impact on student performance. This influence is seen very clearly in the scatterplot of the correlation between traditional EGMA *Addition Level 1* scores and SA-EGMA *Addition Level 1* scores in **Exhibit 14**.

Exhibit 14. Pearson's Correlation Graph for Concurrent Validity, *Addition Level 1* Task



There is a ceiling of scores across the top, showing a large percentage of students scoring highly on the self-administered task (vertical axis), while performing across the range of scores in the traditional task (horizontal axis). This shows that, for a large portion of students, self-administered *Addition Level 1* scores do not accurately reflect traditional EGMA scores, and students tend to score higher on the self-administered than the traditional task. We believe this is due to the difference in underlying constructs created by the removal of time limits from the SA-EGMA tasks. In a traditional EGMA, the *Addition Level 1* task is designed to measure fluency and automaticity by asking the student to complete as many items as possible in 60 seconds. Because it is too difficult to disentangle the time it takes a child to solve an *Addition* task and the time it takes for them to record their answer and move on to the next task in the self-assessment format, we cannot sufficiently measure these constructs in the SA-EGMA and are therefore only measuring knowledge (accuracy) of basic mathematics skills. This difference between the constructs measured by these two assessments is more apparent in contexts where children have lower levels of numerical fluency and automaticity but have still mastered some basic mathematics skills. As a result, we do not recommend using the *Addition* or *Subtraction* tasks to predict traditional EGMA scores, although they function well as part of the overall SA-EGMA.

Because this slightly lower concurrent validity has been seen on more than one iteration of the SA-EGMA, and because we can now see the different constructs being measured by the two tools in this context, we also recommend reducing the number of items in the *Addition Level 1* and *Subtraction Level 1* tasks to the eight items we have reviewed in this report. More items are not necessary for measuring knowledge of these skills, compared with measuring in the traditional assessment. We can reliably assess a student's knowledge of numeracy skills with fewer items in the task, allowing for a shorter assessment. We believe the SA-EGMA *Addition* and *Subtraction* tasks are valid measures of numeracy skills and can be used for assessment, but we no longer need so many items.

We can recommend using the *Missing Number*, *Number Discrimination*, and *Word Problems* tasks as stand-alone tasks for assessing student's emerging numeracy and number sense, and we recommend the overall SA-EGMA instrument as a valid and reliable tool for assessing numeracy skills in a population (group).

5 Conclusions and Recommendations

The primary aim of the effort that culminated in this report was to develop tablet-based assessments that Grade 1-4 students in Tanzania could self-administer and that would effectively and reliably assess their foundational literacy and numeracy skills. We strove to develop instruments that would have high concurrent validity with the well-known and widely used "traditional" EGRA and EGMA. We believe the effort has been successful.

5.1 Literacy

The internal consistency of the SA-EGRA instrument as an assessment, and within each of the individual tasks, is strong and points to consistent tasks and a cohesive assessment of literacy. Analysis of test-retest reliability shows that the SA-EGRA is also a reliable measure of Kiswahili literacy in Grades 1-4 in Tanzania. The SA-EGRA also shows strong construct validity, when the Composite Scores are correlated with traditional ORF scores, even though it cannot predict them at the individual level. As such, the scores from this instrument can be used to create generalized traditional ORF estimates at the population (group) level. We additionally recommend a review of the Composite Scores weighting and, if necessary, adjusting the weights to better reflect the specific context of literacy acquisition in this population. This

Kiswahili SA-EGRA is a reliable, valid instrument for assessing Early Grade Literacy in Grades 1-4 students in Tanzania.

Recommendations Before Next Use with MsingiTek:

- Remove 2–4 *Spelling* items that are performing similarly in terms of Difficulty and Discrimination, or which target similar skills. Specific analysis to inform such choices is in **Annex B.**
- Remove 6–10 *Vocabulary* items that are performing similarly in terms of Difficulty and Discrimination, or which target similar skills. Specific analysis to inform such choices is in **Annex B.**
- Enhance supervising adult training to instruct students in methods to move on if they are stuck on any one item or task.

Recommendations Before Next General Use:

- When focused on assessing only a particular grade or literacy task, shorten the assessment to the needed task(s) for that grade or curriculum, to prevent test fatigue.
- Review the composite score weighting for the tasks with Kiswahili literacy experts and adjust the weights if needed for the specific context.

5.2 Numeracy

The SA-EGMA instrument was found to be internally consistent as an assessment with the factor analysis of the task percent scores showing high factor loadings on a single factor for all tasks. However, within the tasks, many items had factor loadings lower than ideal although not outside the realm of traditional EGMA results. Test-retest analysis found the SA-EGMA to be a reliable assessment across timepoints, with students scoring very similarly. The validity of the SA-EGMA was generally very good, with most tasks correlating highly with traditional EGMA tasks assessing the same constructs. The concurrent validity analysis made clear that the traditional EGMA and the SA-EGMA are measuring different constructs in the *Addition* and *Subtraction* tasks, without the ability to time students to measure fluency and automaticity in the self-administered version. Because of this, we do not recommend using the SA-EGMA *Addition* and *Subtraction* tasks to predict traditional EGMA scores. However, we believe the overall SA-EGMA instrument is a valid method to assess knowledge of numeracy skills. We also recommend *Missing Number* and *Word Problems* as valid and reliable stand-alone measures of number sense and basic numeracy skills.

Recommendations Before Next Use with MsingiTek:

- Enhance supervising adult training to better orient students on the tablet-based assessment format to reduce students' learning curve and reduce response errors in the *Number Identification Subtask*.

Recommendations Before Next General Use:

- When it is possible, shorten the assessment to only the needed tasks for the grade and curriculum to be assessed to prevent test fatigue.

In summary, the SA-EGRA and SA-EGMA instruments' overall performance was very encouraging. Our results presented here support a conclusion that it is appropriate to deploy the instruments with the minor modifications recommended here for assessing Kiswahili literacy and numeracy in early grades in Tanzania. As with any assessment tool, we recommend seeking further opportunities to use and iteratively revise them as evidence accumulates about their performance in various contexts.

Annex A. Addition and Subtraction Level 1 Analysis

This annex reviews the full suite of items in the initial *Addition Level 1* and *Subtraction Level 1* tasks. Items 2, 4, 6, 9, and 10 were cut from each, to leave each task with only eight items. As the purpose of the task from the traditional EGRA has changed in the SA-EGRA to no longer measure fluency, multiple items with identical or similar specifications (measuring the same sub-skill) do not need to be included. Other factor analysis and Pearson's correlation analysis presented in the report use the reduced task, so the analysis with the full suite of items is presented here in **Exhibits A1–A4**.

Exhibit A1. Item Factor Analysis for the Addition Levels 1 and 2 Tasks

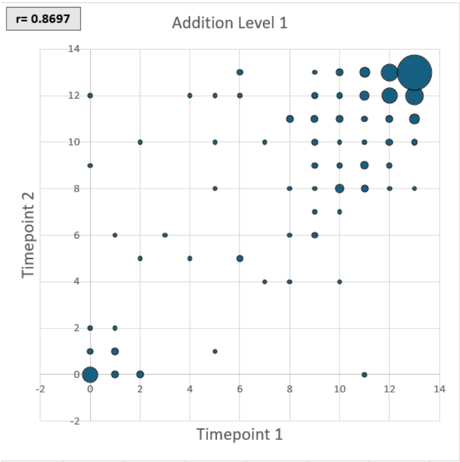
Item	Label	Factor Analysis
Addition 1	1 + 3 = (4)	0.4016
Addition 2	2 + 3 = (5)	0.2900
Addition 3	6 + 2 = (8)	0.4264
Addition 4	4 + 5 = (9)	0.1630
Addition 5	3 + 3 = (6)	0.1548
Addition 6	7 + 3 = (10)	0.5488
Addition 7	8 + 1 = (9)	0.4970
Addition 8	2 + 8 = (10)	0.4704
Addition 9	7 + 5 = (12)	0.3558
Addition 10	8 + 6 = (14)	0.2732
Addition 11	9 + 8 = (17)	0.4230
Addition 12	10 + 2 = (12)	0.2858
Addition 13	8 + 10 = (18)	0.4128
Add Lvl2 1	13 + 6 = (19)	0.2065
Add Lvl2 2	18 + 7 = (25)	0.1808
Add Lvl2 3	12 + 14 = (26)	0.2850
Add Lvl2 4	22 + 37 = (59)	0.4429
Add Lvl2 5	38 + 26 = (64)	0.3694

Exhibit A2. Item Factor Analysis for the Subtraction Levels 1 and 2 Tasks

Item	Label	Factor Analysis
Subtraction 1	4 – 3 = (1)	0.2508
Subtraction 2	5 – 3 = (2)	0.2410
Subtraction 3	8 – 2 = (6)	0.1536

Item	Label	Factor Analysis
Subtraction 4	9 – 5 = (4)	0.2442
Subtraction 5	6 – 3 = (3)	0.0481
Subtraction 6	10 – 3 = (7)	0.2112
Subtraction 7	9 – 1= (8)	0.1078
Subtraction 8	10 – 8 = (2)	0.0343
Subtraction 9	12 – 5 = (7)	0.1873
Subtraction 10	14 – 6 = (8)	0.1866
Subtraction 11	17 – 8 = (9)	0.4585
Subtraction 12	12 – 2 = (10)	0.3673
Subtraction 13	18 – 10 = (8)	0.3825
Sub Lvl2 1	19 – 6 = (13)	0.1704
Sub Lvl2 2	25 – 7 = (18)	0.2978
Sub Lvl2 3	26 – 14 = (12)	0.4708
Sub Lvl2 4	59 – 37 = (22)	0.3986
Sub Lvl2 5	64 – 26 = (38)	0.4913

Exhibit A3. Pearson's Correlation Generalized Test-Retest Reliability for the Addition and Subtraction Level 1 Tasks



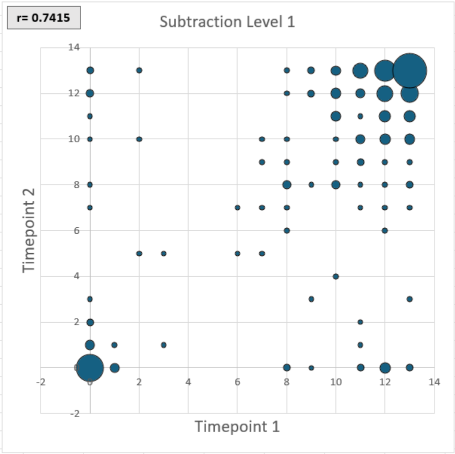
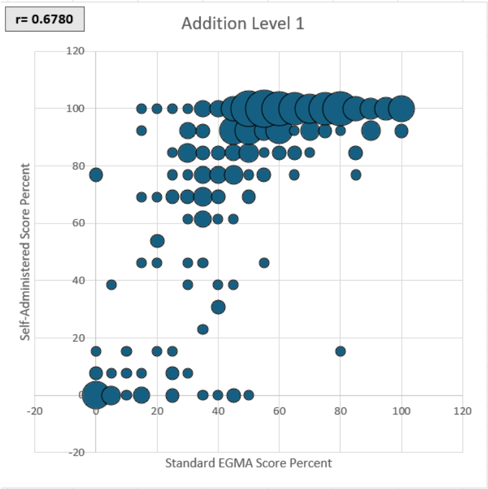
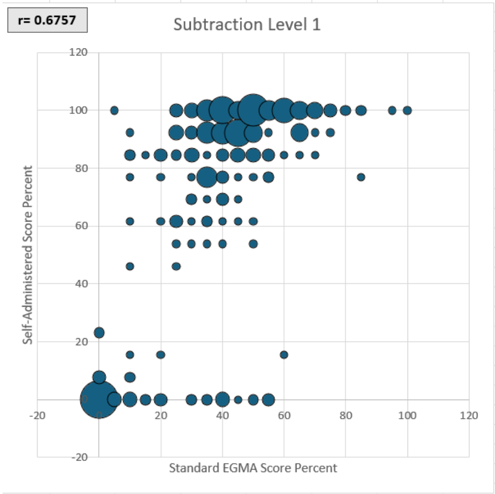


Exhibit A4. Pearson's Correlation Generalized Concurrent Validity of SA-EGMA vs. Paper-Based for Addition and Subtraction Level 1 Tasks





Annex B. IRT Analysis of Spelling and Vocabulary Task Items to Support Item Reduction

Spelling and *Vocabulary* task items were analyzed by an RTI psychometrician specifically to look at candidates for removal that may have similar difficulty or discrimination. The *Spelling* items were recoded using a polytomous, 4-point scale, to allow for comparison across words of different lengths. The *Vocabulary* items are analyzed in their original correct/incorrect form. This annex provides the outcome of that analysis for use in reducing the items of these tasks. This analysis should be used in combination with specific contextual expertise of literacy in Tanzanian Kiswahili to remove items from the assessment for time reduction.

Analytic Caution

While items below are flagged as candidates for removal, once a final item removal list is established, psychometric analyses should be conducted again without these items to establish the impact on measurement efficacy. There can never be a guarantee that the revised item set will yield identical psychometrics on the next data collection after item removal. However, a preliminary analysis can help inform potential future measurement capabilities.

Vocabulary

Discrimination & Misfit: There is a single *Vocabulary* item (vocab2) that has very low discrimination and large misfit (although they often go together). However, there are a couple other items that have lower than ideal discrimination (vocab20 & vocab 5).

Item Difficulty Overlap: There are several *Vocabulary* items that have the same (or closely estimated) item difficulty. These items indicate the potential for redundant measurement, and it is possible that one could be removed without greatly impacting the overall measurement properties of the whole scale. The estimates provided below are on the logit scale and range from 2.4 (vocab2 – most difficult item) to -0.9 (vocab13 – easiest item).

- Vocab6 & vocab20 (difficulty = -0.18)
- Vocab1 & vocab11 (difficulty = -0.2)
- Vocab15 & vocab16 (difficulty = -0.31)
- Vocab7 & vocab18 (difficulty = -0.33)

Spelling

Discrimination & Misfit: While none of the *Spelling* items show large misfit statistics, there are a couple items that have lower than ideal discrimination (spelling_1 & spelling_2).

Item Difficulty Overlap: There are only two *Spelling* items that show difficulty overlap, or stacking. Similar to the *Vocabulary* items, the estimates below are on the logit scale and range from 0.64 (hardest) to -0.81 (easiest).

- Spelling_2 & spelling_10 (difficulty = 0.33)

Summary

Exhibit B1 provides similar information to the overall analysis above and provides indications of items that may be candidates for deletion based on low discrimination, high misfit, and difficulty similarity or stacking.

Exhibit B1. Items for Possible Removal

Item	Discrimination	Misfit	Stacking
spelling_1	X		
spelling_2	X		X
spelling_10			X
vocab1			X
vocab2	X	X	
vocab6			X
vocab11			X
vocab15			X
vocab16			X
vocab20			X

Annex C. SA-EGRA and SA-EGMA Detailed Internal Consistency Analysis

Detailed Internal Consistency Analysis SA-EGRA

Exhibit C1. Factor Analysis Loadings for SA-EGRA Task Scores

SA-EGRA Task Percent Score	Factor 1 Loadings
Letter Sounds	0.4916
Syllables	0.6755
Short Story Reading Comprehension	0.7597
Silent Reading Comprehension	0.7957
Vocabulary	0.8465
Syntax	0.8186
Spelling	0.8067

Exhibit C2. Item Factor Analysis and IRT for the Letter Sounds Task

		Item Response Theory		
Item Number	Factor Analysis	Discrimination	Difficulty	Bi-serial Correlation
1	0.5605	0.4	0.86	0.61
2	0.7100	0.36	0.89	0.76
3	0.7984	0.76	0.65	0.6
4	0.6394	0.59	0.82	0.81
5	0.6965	0.33	0.9	0.72
6	0.5249	0.52	0.84	0.8
7	0.8007	0.6	0.79	0.69
8	0.6826	0.4	0.87	0.72
9	0.6632	0.4	0.85	0.58
10	0.7398	0.52	0.84	0.81
11	0.5133	0.55	0.83	0.72
12	0.7949	0.35	0.89	0.68

Exhibit C3. Item Factor Analysis and IRT for the Vocabulary Task

		Item Response Theory		
Item Number	Factor Analysis	Discrimination	Difficulty	Bi-serial Correlation
1	0.7225	0.83	0.71	0.73
2	0.3778	0.78	0.7	0.7
3	0.7716	0.67	0.71	0.57
4	0.6770	0.82	0.55	0.65
5	0.4375	0.52	0.8	0.57
6	0.6307	0.62	0.78	0.62
7	0.6740	0.6	0.72	0.55
8	0.5040	0.64	0.72	0.63
9	0.6836	0.76	0.6	0.63
10	0.6937	0.73	0.73	0.65
11	0.5493	0.73	0.61	0.6
12	0.6180	0.58	0.36	0.44
13	0.5404	0.55	0.7	0.48
14	0.6169	0.79	0.73	0.77
15	0.5035	0.74	0.74	0.69
16	0.6046	0.55	0.69	0.49
17	0.6011	0.71	0.71	0.64
18	0.6392	0.73	0.73	0.69
19	0.5646	0.63	0.67	0.55
20	0.4282	0.77	0.69	0.69

Exhibit C4. Item Factor Analysis and IRT for the Syntax Task

		Item Response Theory		
Item Number	Factor Analysis	Discrimination	Difficulty	Bi-serial Correlation
1	0.7273	0.9	0.63	0.79
2	0.5949	0.61	0.81	0.66
3	0.5994	0.79	0.6	0.67
4	0.6743	0.83	0.72	0.7
5	0.6600	0.85	0.61	0.72

		Item Response Theory		
Item Number	Factor Analysis	Discrimination	Difficulty	Bi-serial Correlation
6	0.5819	0.81	0.65	0.63
7	0.6210	0.63	0.82	0.66
8	0.5942	0.71	0.7	0.67

Exhibit C5. Item Factor Analysis and IRT for the Syllables Task

		Item Response Theory		
Item Number	Factor Analysis	Discrimination	Difficulty	Bi-serial Correlation
1	0.5106	0.61	0.74	0.6
2	0.6883	0.72	0.76	0.75
3	0.6544	0.76	0.57	0.47
4	0.5325	0.43	0.86	0.7
5	0.5330	0.28	0.92	0.66
6	0.6603	0.51	0.8	0.59
7	0.6429	0.6	0.76	0.6
8	0.4903	0.51	0.84	0.69
9	0.7094	0.37	0.89	0.67
10	0.7087	0.48	0.82	0.56
11	0.3347	0.43	0.87	0.71

Exhibit C6. Item Factor Analysis and IRT for the Short Reading Comprehension Task

		Item Response Theory		
Item Number	Factor Analysis	Discrimination	Difficulty	Bi-serial Correlation
1	0.4143	0.64	0.71	0.57
2	0.4325	0.79	0.62	0.59
3	0.6462	0.84	0.65	0.7
4	0.7140	0.88	0.66	0.76
5	0.6741	0.78	0.75	0.74
6	0.5436	0.8	0.62	0.65

Exhibit C7. Item Factor Analysis and IRT for the Silent Reading Comprehension Task

		Item Response Theory		
Item Number	Factor Analysis	Discrimination	Difficulty	Bi-serial Correlation
1	0.4818	0.83	0.53	0.59
2	0.4716	0.47	0.87	0.54
3	0.4293	0.63	0.77	0.53
4	0.6089	0.9	0.52	0.68
5	0.4191	0.76	0.67	0.54
6	0.5568	0.71	0.63	0.63
7	0.4805	0.67	0.72	0.57
8	0.4551	0.71	0.65	0.56

Exhibit C8. Item Factor Analysis for the Spelling Task

Item	Factor Analysis
1	0.6648
2	0.7162
3	0.8442
4	0.7600
5	0.8734
6	0.8507
7	0.8382
8	0.7948
9	0.8549
10	0.8594
11	0.8957
12	0.8252

Detailed Internal Consistency SA-EGMA

Exhibit C9. Factor Analysis Loadings for SA-EGMA Task Scores

Task Percent Score	Factor 1 Loadings
Number Identification	0.6268
Number Discrimination	0.7203
Missing Number	0.7732
Addition	0.6738
Addition Level 2	0.7926
Subtraction	0.6403
Subtraction Level 2	0.7269
Word Problems	0.6958

Exhibit C10. Item Factor Analysis for the Number Identification Task

Item	Factor Analysis
1	0.0527
2	-0.0528
3	-0.2459
4	0.108
5	0.3117
6	0.3262
7	0.5655
8	0.2004
9	0.3641
10	0.4848
11	0.6192
12	0.6479

Exhibit C11. Item Factor Analysis for the Number Discrimination Task

Item	Factor Analysis
1	0.5342
2	0.3846
3	0.4342
4	0.4784
5	0.463
6	0.3289
7	0.6259
8	0.3052
9	0.4692
10	0.4422

Exhibit C12. Item Factor Analysis for the Missing Number Task

Item	Factor Analysis
1	0.1995
2	0.2305
3	0.3351
4	0.1405
5	0.6726
6	0.1454
7	0.5891
8	0.5259
9	0.4096
10	0.5457

Exhibit C13. Item Factor Analysis for the Addition Level 1 and 2 Tasks

Item	Factor Analysis
1	0.3862
2	0.4762
3	0.1876
4	0.5249
5	0.3072
6	0.4009
7	0.3515
8	0.1379
9	0.1718
10	0.3149
11	0.4257
12	0.3777
13	0.3862

Exhibit C14. Item Factor Analysis for the Subtraction Level 1 and 2 Tasks

Item	Factor Analysis
1	0.1718
2	0.1974
3	0.0605
4	0.0186
5	-0.0567
6	0.4139
7	0.4187
8	0.1717
9	0.3071
10	0.5005
11	0.4066
12	0.4762
13	0.1718

Exhibit C15. Item Factor Analysis for the Word Problems Task

Item	Factor Analysis
1	0.3578
2	0.1420
3	0.4667
4	0.2331
5	0.4014
6	0.5948

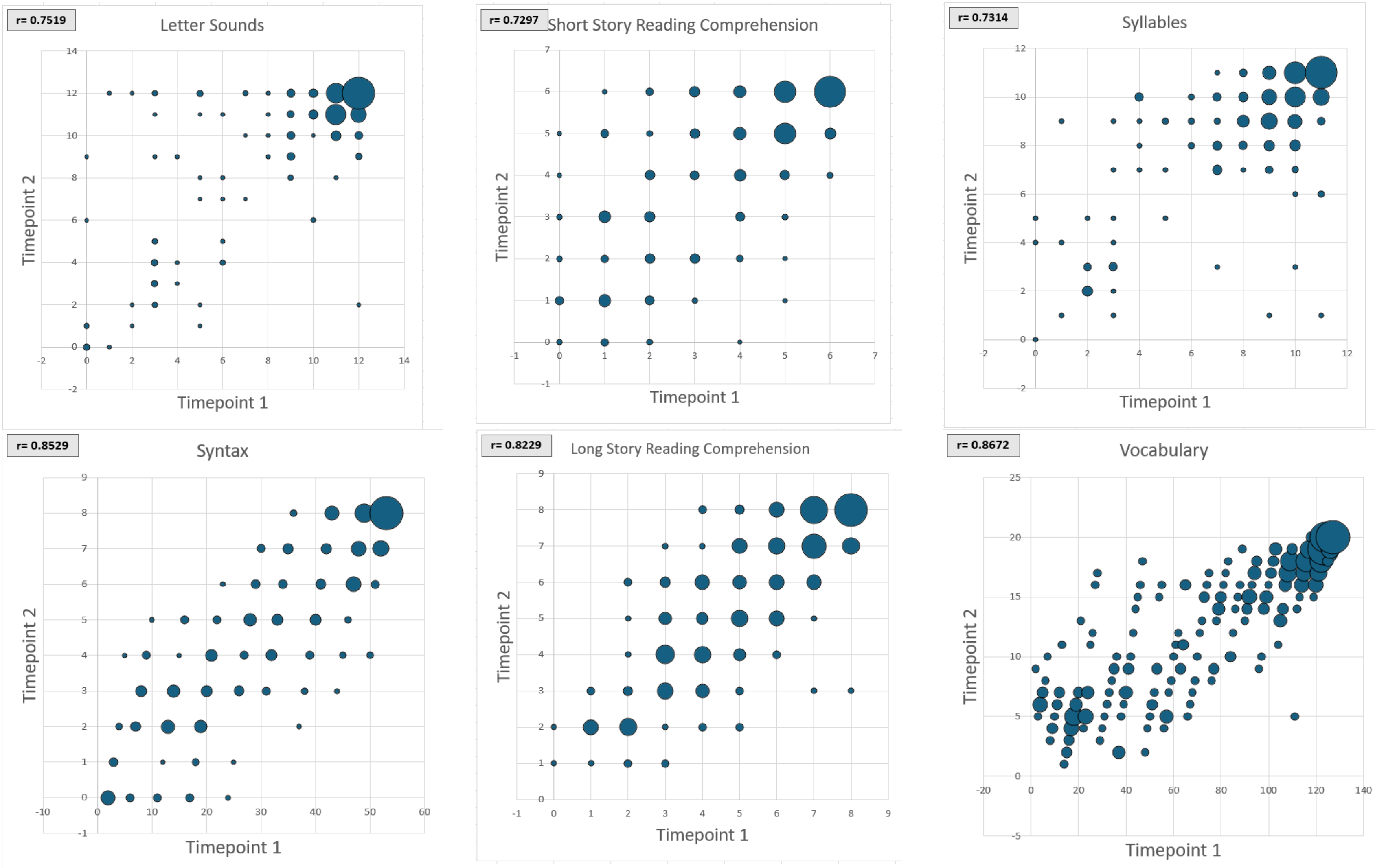
Annex D. Test-Retest Reliability: Pearson's Correlation and Bland-Altman Plots

Detailed Test-Retest Reliability Analysis SA-EGRA

Exhibit D1. Pearson's Correlation for the SA-EGRA Test-Retest

SA-EGRA Task Percent Score	Correlation
Letter Sounds	0.7519
Short Reading Comprehension	0.7297
Syllables	0.7314
Syntax	0.8529
Long Reading Comprehension	0.8229
Vocabulary	0.8672
Spelling	0.9234

Exhibit D2. SA-EGRA Pearson's Correlation Generalized Test-Retest Reliability, by Task



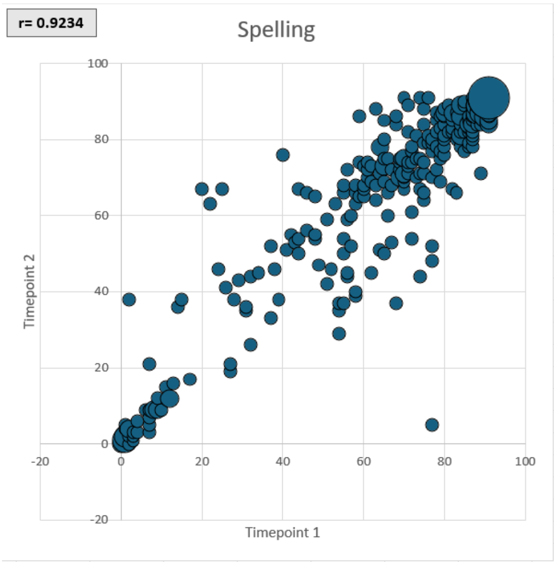
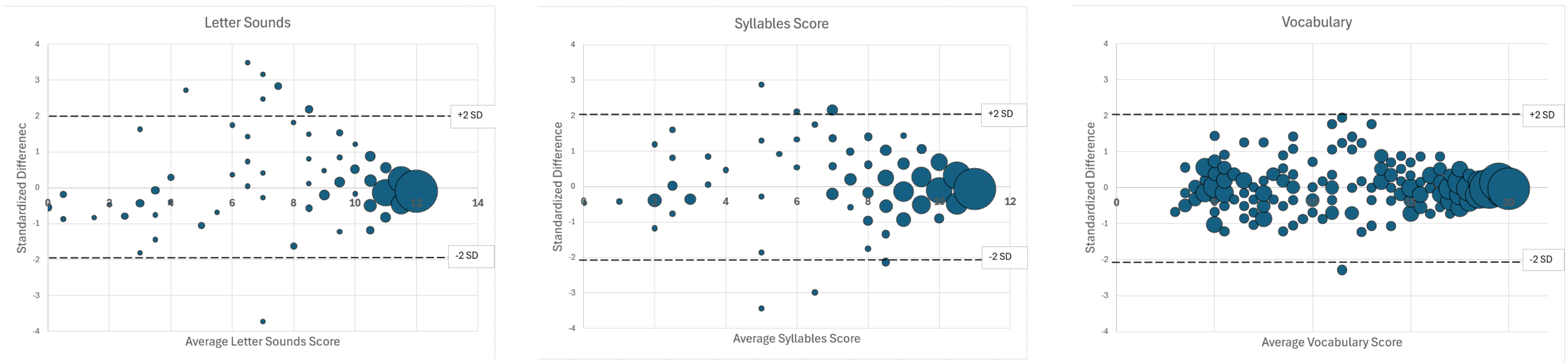
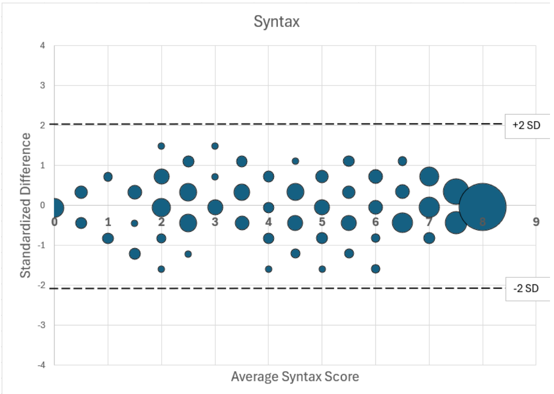
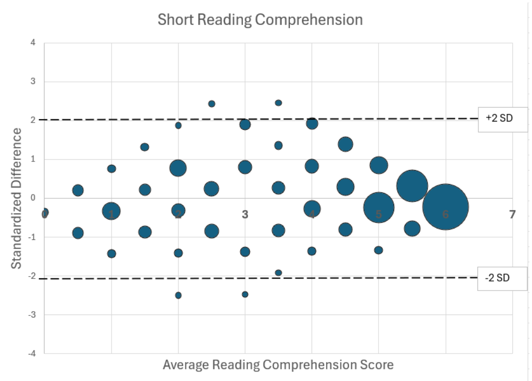
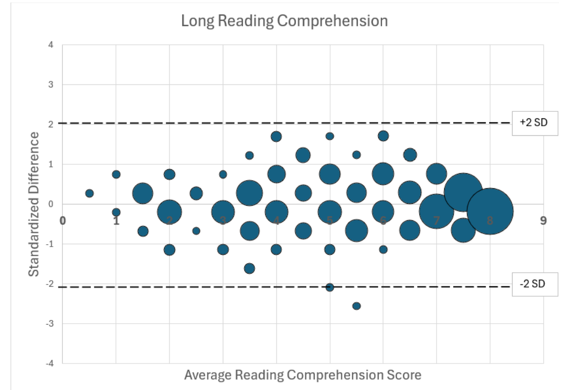


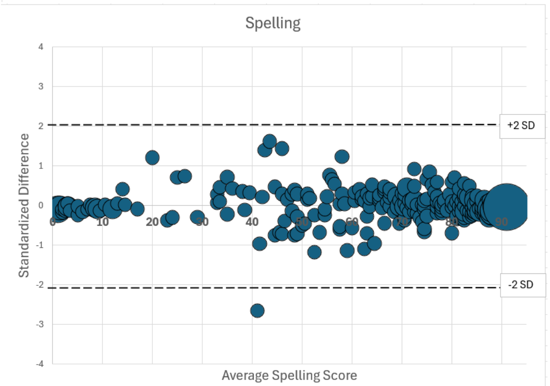
Exhibit D3. SA-EGRA Bland-Altman Plots of Test-Retest Reliability, by Task




Syntax
Standardized Difference
+2 SD
-2 SD
Average Syntax Score


Short Reading Comprehension
Standardized Difference
+2 SD
-2 SD
Average Reading Comprehension Score


Long Reading Comprehension
Standardized Difference
+2 SD
-2 SD
Average Reading Comprehension Score

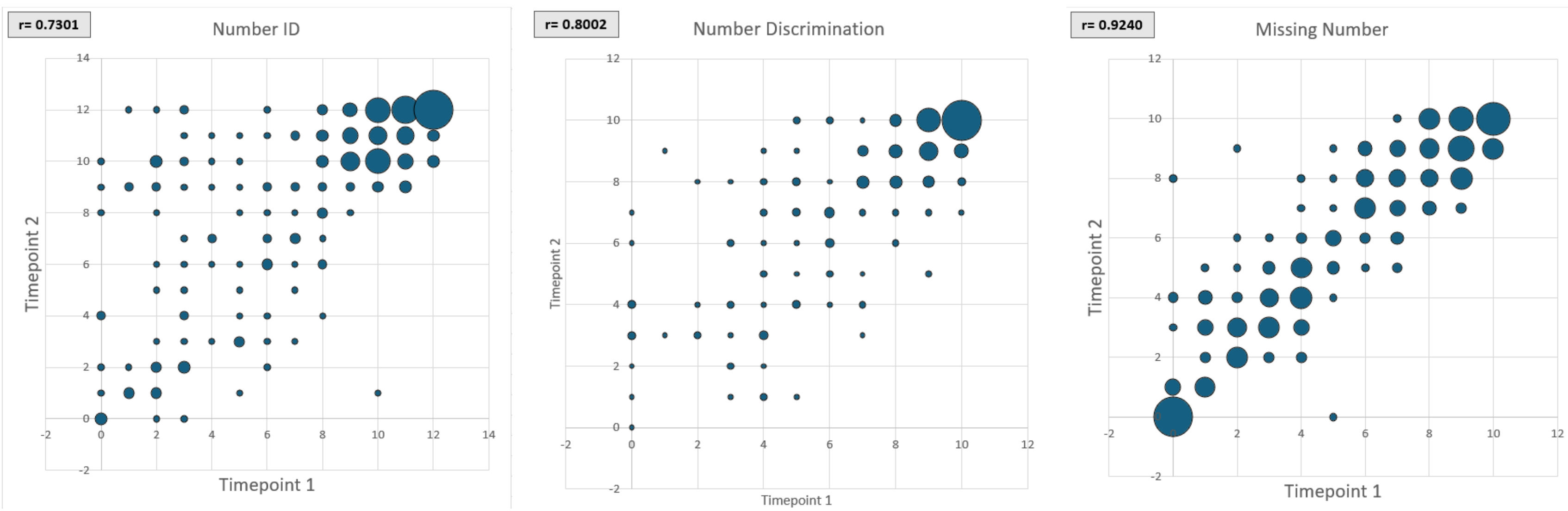

Spelling
Standardized Difference
+2 SD
-2 SD
Average Spelling Score

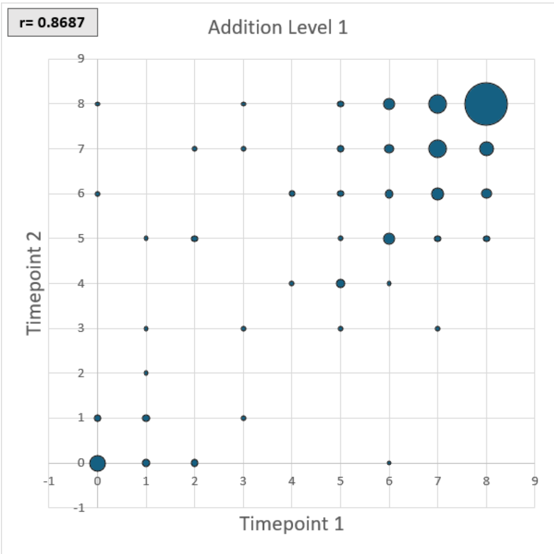
Detailed Test-Retest Reliability Analysis SA-EGMA

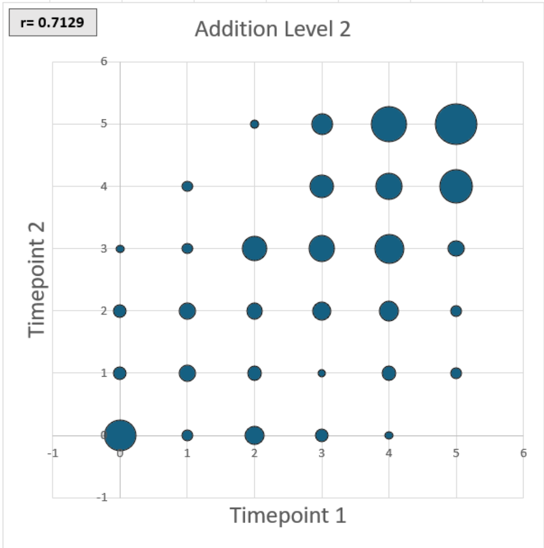
Exhibit D4. Pearson's Correlation for the SA-EGMA Test-Retest

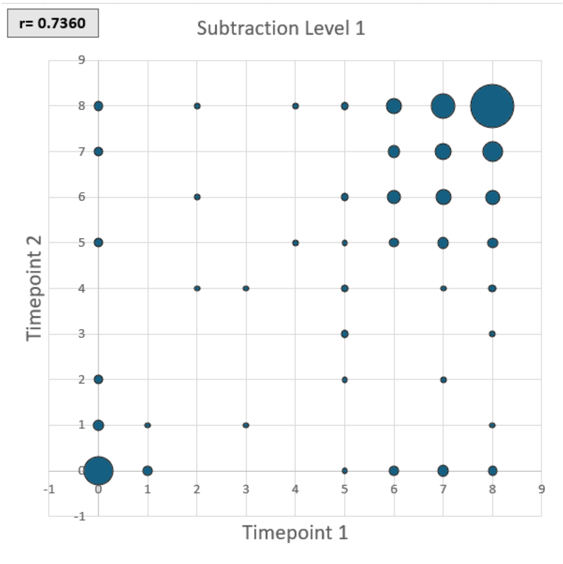
SA-EGRA Task Percent Score	Correlation
Number Identification	0.7301
Number Discrimination	0.8002
Missing Number	0.9240
Addition Level 1	0.8687
Addition Level 2	0.7129
Subtraction Level 1	0.7360
Subtraction Level 2	0.6377
Word Problems	0.7880

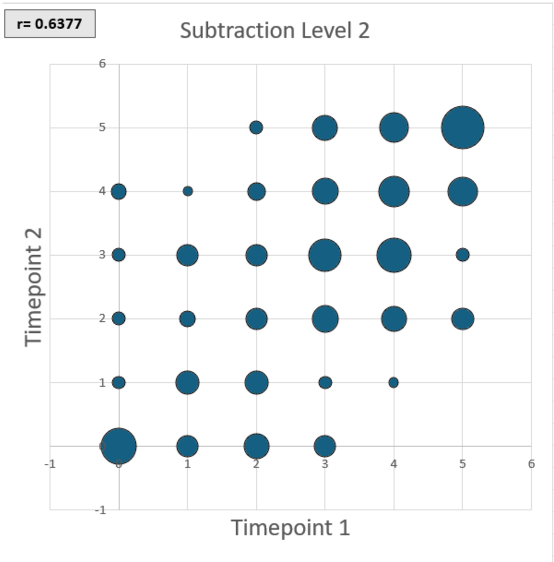
Exhibit D5. SA-EGMA Pearson's Correlation Generalized Test-Retest Reliability, by Task




r= 0.8687
Addition Level 1
Timepoint 2
Timepoint 1


r= 0.7129
Addition Level 2
Timepoint 2
Timepoint 1


r= 0.7360
Subtraction Level 1
Timepoint 2
Timepoint 1


r= 0.6377
Subtraction Level 2
Timepoint 2
Timepoint 1

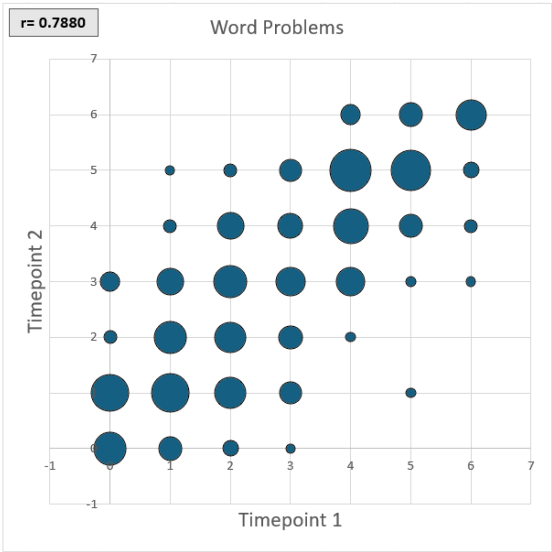
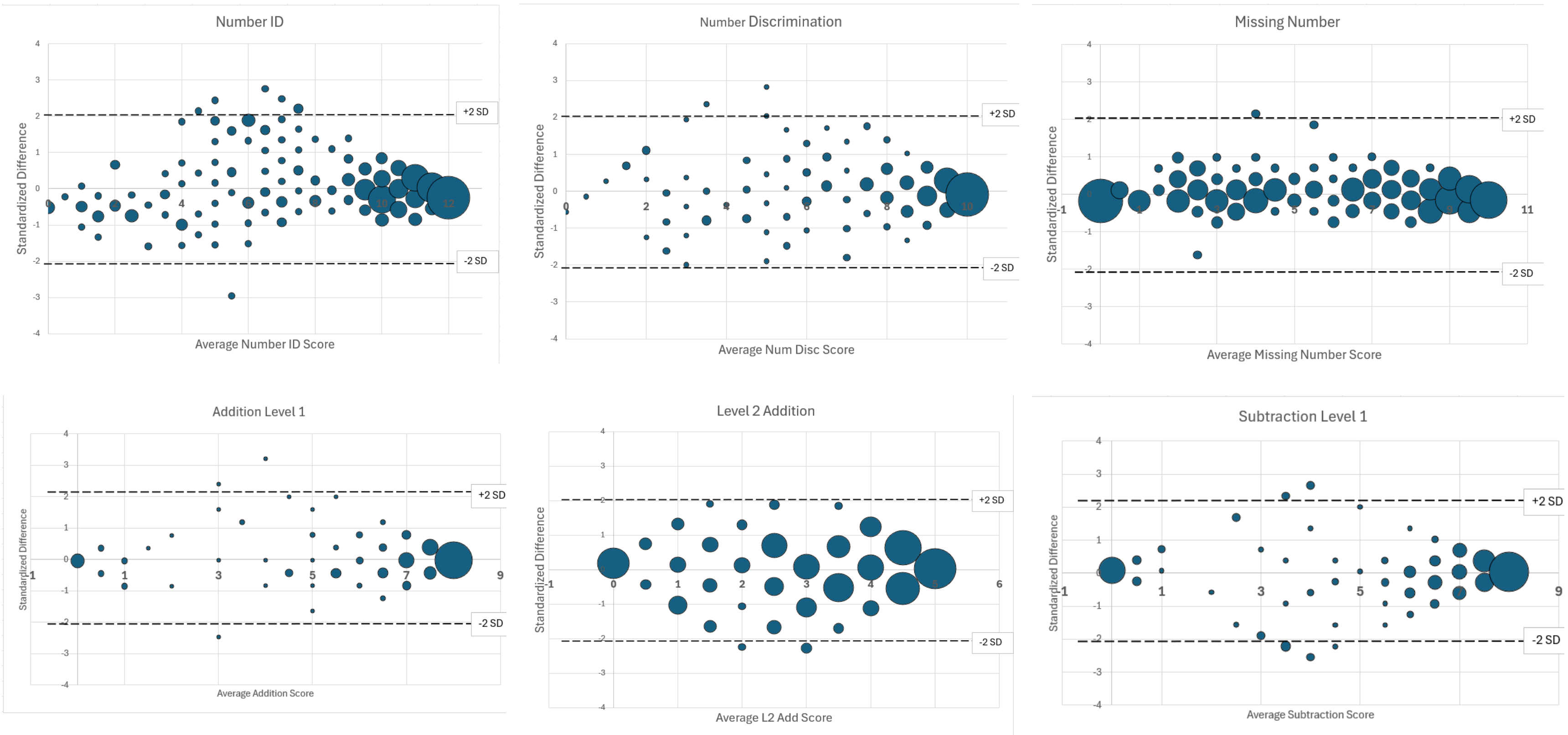
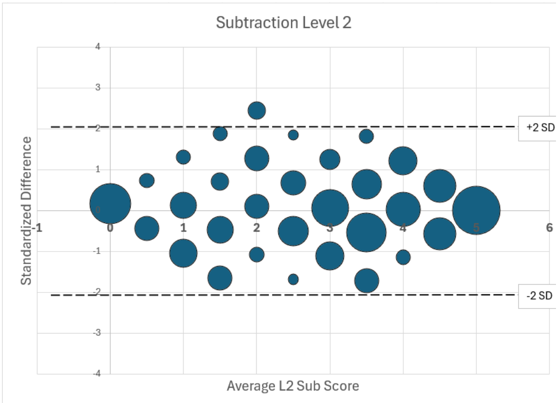
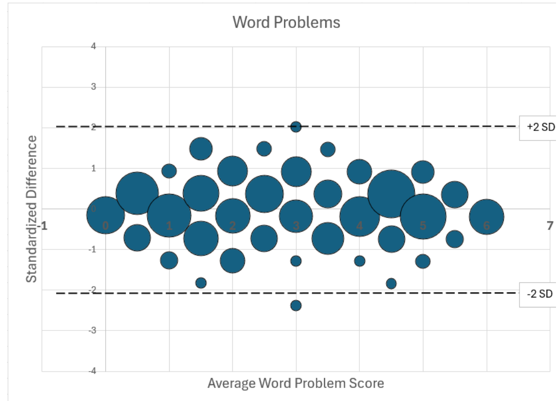

r= 0.7880
Word Problems
Timepoint 2
Timepoint 1

Exhibit D6. SA-EGMA Bland-Altman Plots of Test-Retest Reliability, by Task




Subtraction Level 2
Standardized Difference
Average L2 Sub Score
+2 SD
-2 SD


Word Problems
Standardized Difference
Average Word Problem Score
+2 SD
-2 SD

Annex E. SA-EGRA Construct Validity and SA-EGMA Concurrent Validity

Detailed Construct and Concurrent Validity Analysis SA-EGRA

Exhibit E1. SA-EGRA Spelling Score Percent score vs. Traditional EGRA Oral Reading Fluency

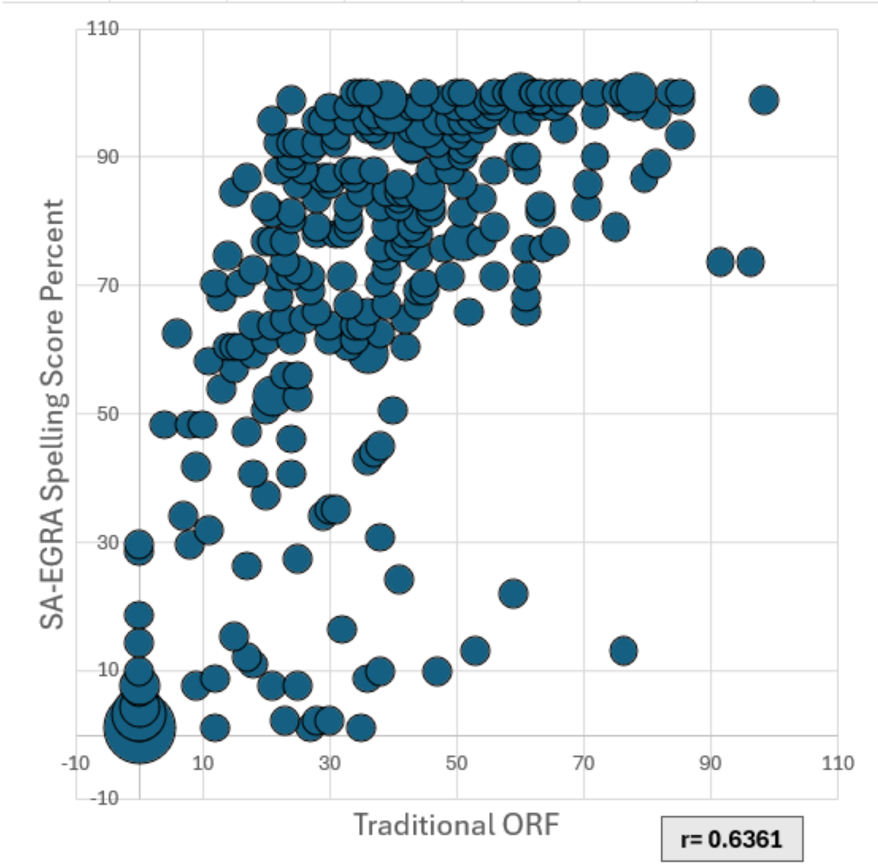
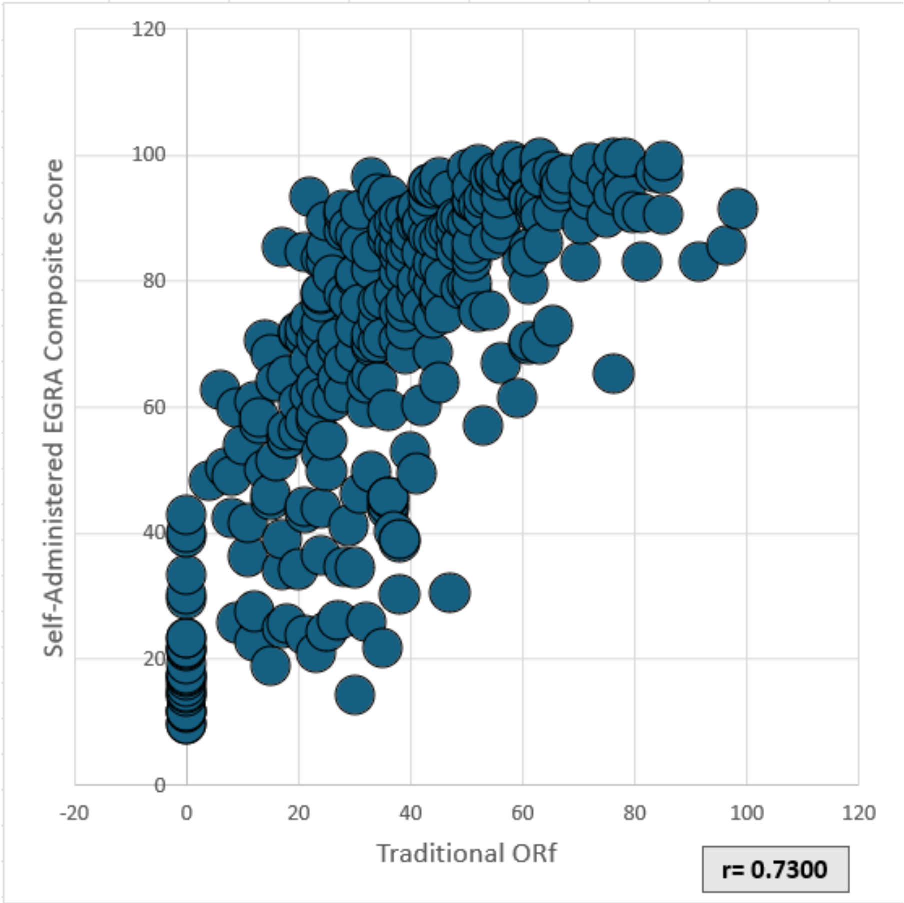


Exhibit E2. SA-EGRA Composite Score Percent score vs. Traditional EGRA Oral Reading Fluency



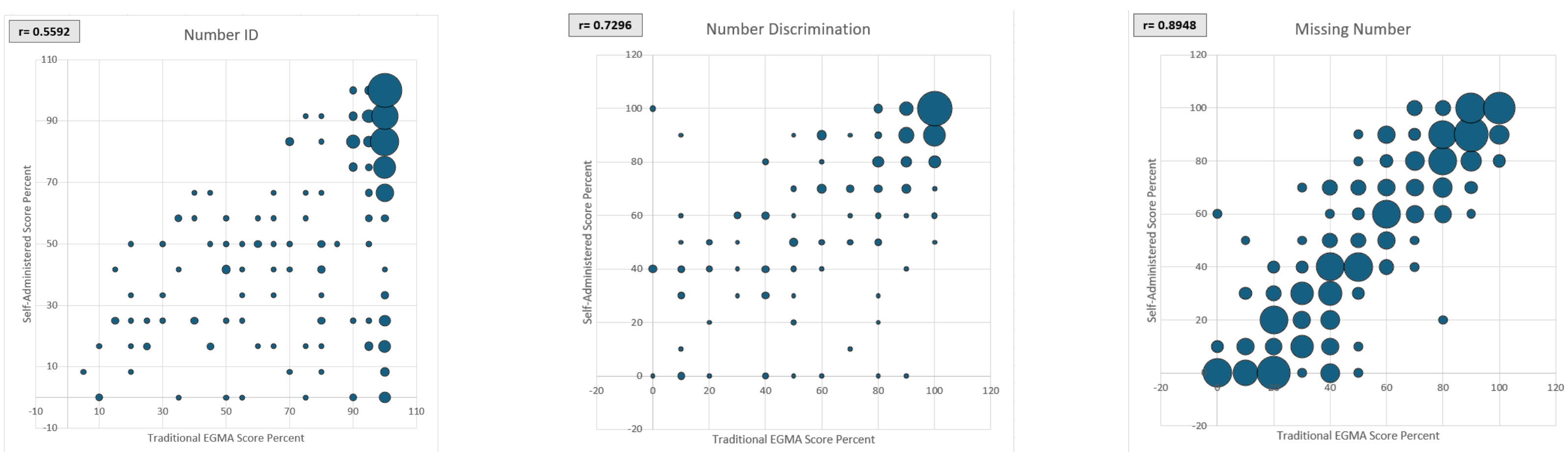
Composite score calculation: SA_EGRA_composite = 0.4 × spelling_total_score_pcnt + 0.15 × short_read_comp_score_pcnt + 0.1 × long_read_comp_score_pcnt + 0.05 × letter_sounds_score_pcnt + 0.1 × vocab_score_pcnt + 0.1 × syntax_score_pcnt + 0.1 × syllables_score_pcnt

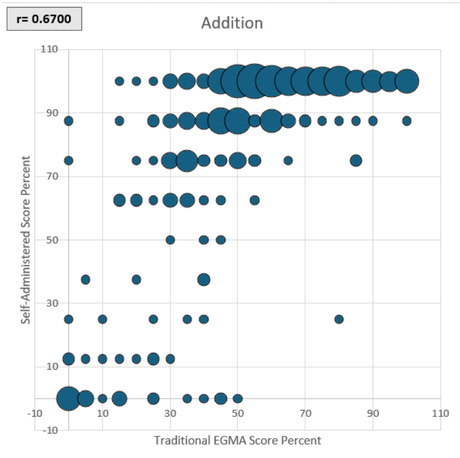
Detailed Construct and Concurrent Validity Analysis SA-EGMA

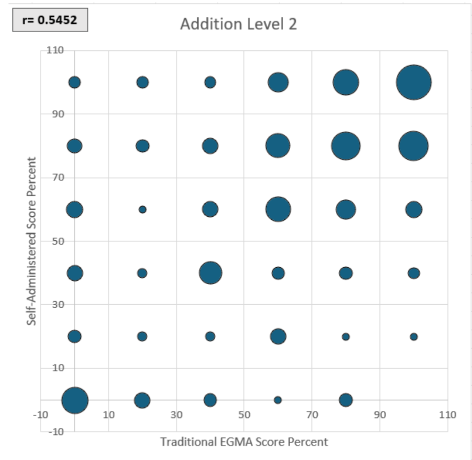
Exhibit E3. Pearson's Correlation for Generalized Concurrent Validity of the SA-EGMA and Traditional EGMA

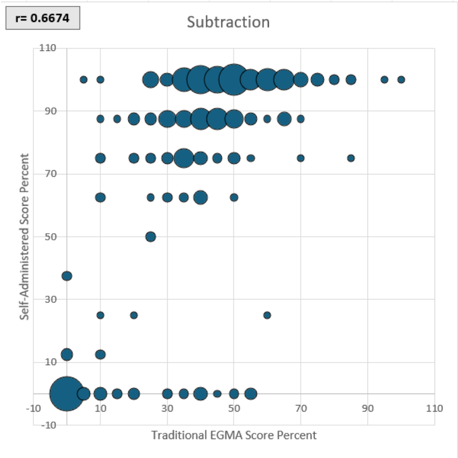
SA-EGMA Task Percent Score	Correlation
Number Identification	0.5592
Number Discrimination	0.7296
Missing Number	0.8948
Addition Level 1	0.6700
Addition Level 2	0.5452
Subtraction Level 1	0.6674
Subtraction Level 2	0.5358
Word Problems	0.7296

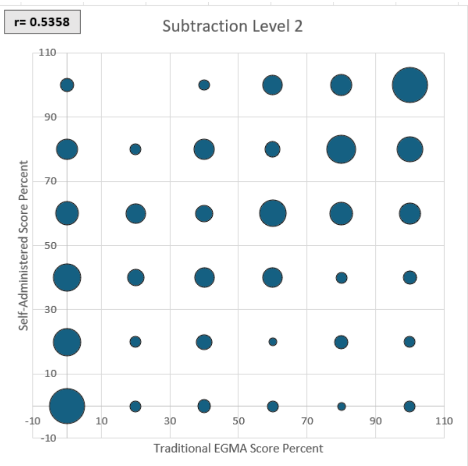
Exhibit E4. Pearson's Correlation Generalized Concurrent Validity of SA-EGMA vs. Paper-Based

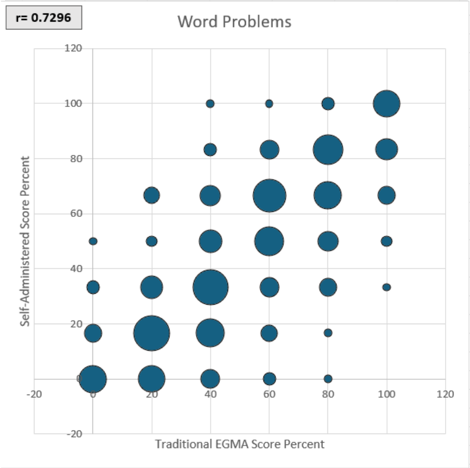



r= 0.6700
Addition
Self-Administered Score Percent
Traditional EGMA Score Percent


r= 0.5452
Addition Level 2
Self-Administered Score Percent
Traditional EGMA Score Percent


r= 0.6674
Subtraction
Self-Administered Score Percent
Traditional EGMA Score Percent


r= 0.5358
Subtraction Level 2
Self-Administered Score Percent
Traditional EGMA Score Percent


r= 0.7296
Word Problems
Self-Administered Score Percent
Traditional EGMA Score Percent

Annex F. SA-EGRA and SA-EGMA Tasks, Items, and Instructions

EGRA

1. Consent

#	English Prompt	Kiswahili Version	Audio Filename
Consent	Hello. My name is teacher ____ and I live in _____. I am here today because my team has developed a learning game for children. We would like your help with that. First, I want to tell you a bit more about myself. My/I ___ [favorite book, etc.]. To help us with this game, you will be given this tablet device to play a little learning game [point to tablet]. The game will ask you to read some letters, numbers, words and a short story as well as do some number games. The game will take about 30 minutes to play. This is NOT a test for school, and it will not affect your grade at school. No one – not your parents, friends, teacher, or head teacher - will know these are your answers. You do not have to participate if you do not wish to. Once we begin, if at any time you want to stop playing the game, that's also all right. Do you have any questions? *[Answer questions, then go to each child and ask]* Do you want to do this activity? *[If a child says 'yes', say]* Great, thank you so much. You can play this game on your own, just tap this button and follow the instructions. [*Make sure the*	Habari. Jina langu ni mwalimu ____ na ninaishi _____. Nipo hapa leo kwa sababu wenzangu wametengeza mchezo wa kujifunza kwa watoto. Tungependa msaada wako kwenye mchezo huu. Kwanza, natakakukuelezea kidogo kuhusu mimi. Mimi ni ___ [kitabu kizuri, n.k.]. Ili kutusaidia kwenye mchezo huu, utapewa kishikwambi kwa ajili ya kucheza mchezo mdogo wa kujifunza [elekeza kwenye kishikwambi]. Katika mchezo huu utaelekezwa usome baadhi ya herufi, namba, maneno, na hadithi fupi pamoja na kucheza michezo ya namba. Mchezo huu utachukua takriban dakika 30 ili kuucheza. Huu SIO mtihani, na mchezo huu hautakuwa na athari yoyote kwenye tathmini ya maendeleo yako shuleni. Hakuna mtu yoyote - iwe ni mzazi wako, rafiki, mwalimu, au mkuu wa shule - atakayejua kuwa haya ni majibu yako. Hakuna ulazima wa kushiriki katika mchezo huu. Na pia, pindi tutakapoanza, ukitaka kuacha kucheza mchezo huu kwa wakati wowote unaweza kuacha, na hilo pia ni sawa kabisa.Je, una swali lolote? [*Jibu maswali, kisha nenda kwa kila mtoto na muulize*:] Ungependa kushiriki katika shughuli hii? [*Kama mtoto atasema 'ndiyo', sema*:]	N/A (This appears on the first screen but is read aloud by the facilitator and not by the tablet.)

	child proceeds to the first element. If a child says 'no', say] It is ok not to participate. You can now return to your [teacher/class/recess…].	Vizuri sana, asante sana. Unaweza kucheza mchezo huu peke yako, hivyo gusa alama hii kisha fuata maelekezo. [*Hakikisha mtoto anaendelea na sehemu ya kwanza ya mchezo. Kama mtoto atasema 'hapana', sema:*] Ni sawa tu kutoshiriki. Sasa unaweza kurudi kwa [mwalimu wako/darasani/mapumziko…].	
Beginning prompt	*Are you ready? Tap the arrow to begin.*	Je, upo tayari? Gusa alama ya mshale kuanza.	ready

2. Letter Sounds

#	English Version	Kiswahili Version	Audio Filename
Intro Letter Sounds	First, we will play a game with letters! You will see some letters like these and hear a sound. Tap the letter that makes the sound you hear. If you need to hear the sound again, tap the fruit. When you finish, tap the arrow to move on. First, let's do a practice! Are you ready? Tap the arrow to begin.	Kwanza, tutacheza mchezo wa herufi! Utaziona herufi kadhaa kama hizi na sikiliza sauti. Gusa herufi inayotoa sauti unayoisikia. Ukihitaji kuisikia tena sauti hiyo, gusa alama ya tunda. Ukimaliza, gusa alama ya mshale ili kuendelea. Kwanza, hebu tufanye mazoezi! Je, upotayari? Kwa kuanza gusa alama ya mshale.	Letters_intro
Letter Sound Practice	Listen: /sa/. Tap the letter (or letters) for the first sound in /sa/. … /sa/.	Sikiliza: /sa/. Gusa herufi ya sauti ya kwanza kwenye silabi /sa/. … /sa/.	Letters_practice_prompt
	That's right!	Ni sahihi kabisa!	letters_right
	Good try!	Umejaribu vizuri sana!	letters_try
	The correct answer is the letter ***s***. The first sound in /sa/ is made by the letter s. Now you will hear another sound. Are you ready? Tap the arrow to begin.	Jibu sahihi ni herufi **s**. Sauti ya kwanza katika silabi /sa/ imeundwa na herufi **s**. Sasa utasikia sauti nyingine. Je, upo tayari? Kwa kuanza gusaalama ya mshale.	letters_practice_answer
Letter Sound Help	Tap the letter that makes the first sound you hear. To hear the sound again, tap the fruit. When you finish, tap the arrow to move on.	Gusa herufi inayotoa sauti ya kwanza unayoisikia. Ili kuisikia tena sauti, gusa alama ya tunda. Ukimaliza, gusa alama ya mshale ili kuendelea.	**letters_help**

Letter Sound Item Prompt 1	Listen: /ka/. Tap the letter for the first sound in /ka/. … /ka/.	Sikiliza: /ka/. Gusa herufi ya sauti ya kwanza katika silabi /ka/. … /ka/.	Letters_1
		/ka/	letters_1_sound
2	Listen: /ma/. Tap the letter or letters for the first sound in /ma/. … /ma/.	Sikiliza: /ma/. Gusa herufi ya sauti ya kwanza katika silabi /ma/ … /ma/.	Letters_2
		/ma/	letters_2_sound
3	Listen: /ta/. Tap the letter or letters for the first sound in /ta/. … /ta/.	Sikiliza: /ta/. Gusa herufi ya sauti ya kwanza katika silabi /ta/ … /ta/.	Letters_3
		/ta/	letters_3_sound
4	Listen: /za/. Tap the letter or letters for the first sound in /za/. … /za/.	Sikiliza: /za/. Gusa herufi ya sauti ya kwanza katika silabi /za/ … /za/.	Letters_4
		/za/	letters_4_sound
5	Listen: /i/. Tap the letter sound /i/. … /i/.	Sikiliza: /i/. Gusa herufi ya sauti /i/ … /i/.	Letters_5
		/i/	letters_5_sound
6	Listen: /ra/. Tap the letter for the first sound in /ra/. … /ra/.	Sikiliza: /ra/. Gusa herufi ya sauti ya kwanza katika silabi /ra/ … /ra/.	Letters_6
		/ra/	letters_6_sound
7	Listen: /ya/. Tap the letter for the first sound in /ya/. … /ya/.	Sikiliza: /ya/. Gusa herufi ya sauti ya kwanza katika silabi /ya/ … /ya/.	Letters_7
		/ya/	letters_7_sound
8	Listen: /fa/. Tap the letter for the first sound in /fa/. … /fa/.	Sikiliza: /fa/. Gusa herufi ya sauti ya kwanza katika silabi /fa/ … /fa/.	Letters_8
		/fa/	letters_8_sound
9	Listen: /ba/. Tap the letter or letters for the first sound in /ba/. … /ba/.	Sikiliza: /ba/. Gusa herufi ya sauti ya kwanza katika silabi /ba/ … /ba/.	Letters_9
		/ba/	letters_9_sound
10	Listen: /o/. Tap the letter sound /o/. … /o/.	Sikiliza: /o/. Gusa herufi ya sauti /o/ … /o/.	Letters_10
		/o/	letters_10_sound

11	Listen: /nga/. Tap the letter or letters for the first sound in /nga/… /nga/.	Sikiliza: /nga/. Gusa herufi za sauti ya kwanza katika silabi /nga/ … /nga/.	Letters_11
		/nga/	letters_11_sound
12	Listen: /cha/. Tap the letter or letters for the first sound in /cha/. … /cha/.	Sikiliza: /cha/. Gusa herufi za sauti ya kwanza katika silabi /cha/ … /cha/.	Letters_12
		/cha/	letters_12_sound

Items:

#	**Answer Options (correct answer box is gray)**				
Practice	b	m	l	**s**	d
1	t	g	**k**	ng	h
2	**m**	n	ng’	mw	w
3	p	**t**	tw	d	th
4	s	v	dh	gh	**z**
5	a	e	**i**	o	u
6	**r**	l	y	w	n
7	l	j	w	h	**y**
8	th	fy	**f**	z	v
9	**b**	p	d	bw	mb
10	a	e	i	**o**	u
11	nd	**ng**	ny	nj	ng’
12	dh	gh	th	sh	**ch**

3. Syllables

#	English Version	Kiswahili Version	Audio Filename
Intro Syllables	First, we will play a game with syllables! You will see some letters like these and hear a syllable. Tap the letters that makes the syllable you hear. If you need to hear the syllable again, tap the fruit. When you finish, tap the arrow to move on. First, let's do a practice! Are you ready? Tap the arrow to begin.	Kwanza, tutacheza mchezo wa silabi! Utaziona baadhi ya herufi kama hizi na kuzisikia silabi zake. Gusa herufi zinazounda silabi unayoisikia. Kama unahitaji kuzisikia tena silabi hizo, gusa alama ya tunda. Ukimaliza, gusa alama ya mshale ili kuendelea. Kwanza, tufanye mazoezi! Je upo tayari? Kwa kuanza, gusa alama ya mshale.	Syllables_intro
Syllable Practice	Listen: /ka/. Tap the syllable /ka/. … /ka/.	Sikiliza: /ka/. Gusa silabi /ka/ … /ka/.	Syllables_intro
	That's right!	Ni sahihi!	syllables_right
	Good try!	Umefanya vizuri!	syllables_try
	The correct answer is /ka/. The letters k, and a spell the syllable /ka/. Now you will hear another syllable. Are you ready? Tap the arrow to begin. -	Jibu sahihi ni /ka/. Herufi **k**, na a zinatamkwa kwa silabi /ka/. Sasa utazisikia silabi nyingine. Je upo tayari? Kwa kuanza gusa alama ya mshale.	Syllables_practice_answer
Syllable Help	Tap the letters that make the syllable you hear. To hear the syllable again, tap the fruit. When you finish, tap the arrow to move on.	Gusa herufi zinazounda silabi unayoisikia. Ili kuzisikia tena silabi hizo, gusa alama ya tunda. Ukimaliza, gusa alama ya mshale ili kuendelea.	syllables_help
Syllable Item Prompt 1	Listen: /la/. Tap the syllable /la/. … /la/.	Sikiliza: /la/. Gusa silabi /la/ … /la/.	Syllables_1
		/la/	syllables_1_sound
2	Listen: /po/. Tap the syllable /po/ … /po/.	Sikiliza: /po/. Gusa silabi /po/ … /po/.	Syllables_2
		/po/	syllables_2_sound
3	Listen: /wa/. Tap the syllable /wa/. … /wa/.	Sikiliza: /wa/. Gusa silabi /wa/ … /wa/.	Syllables_3
		/wa/	syllables_3_sound
4	Listen: /shu/. Tap the syllable /shu/ … /shu/.	Sikiliza: /shu/. Gusa silabi /shu/ … /shu/.	Syllables_4
		/shu/	syllables_4_sound
5	Listen: /ja/. Tap the syllable /ja/ … /ja/.	Sikiliza: /ja/. Gusa silabi /ja/ … /ja/.	Syllables_5

#	English Version	Kiswahili Version	Audio Filename
		/ja/	syllables_5_sound
6	Listen: /ndi/. Tap the syllable /ndi/. … /ndi/.	Sikiliza: /ndi/. Gusa silabi /ndi/ … /ndi/.	Syllables_6
		/ndi/	syllables_6_sound
7	Listen: /kwe/. Tap the syllable /kwe/. … /kwe/.	Sikiliza: /kwe/. Gusa silabi /kwe/ … /kwe/.	Syllables_7
		/kwe/	syllables_7_sound
8	Listen: /swa/. Tap the syllable /swa/. … /swa/.	Sikiliza: /swa/. Gusa silabi /swa/ … /swa/.	Syllables_8
		/swa/	syllables_8_sound
9	Listen: /mbe/. Tap the syllable /mbe/. … /mbe/.	Sikiliza: /mbe/. Gusa silabi /mbe/ … /mbe/.	Syllables_9
		/mbe/	syllables_9_sound
10	Listen: /nywa/. Tap the syllable /nywa/ … /nywa/.	Sikiliza: /nywa/. Gusa silabi /nywa/ … /nywa/.	Syllables_10
		/nywa/	syllables_11_sound
11	Listen: /ng’u/. Tap the syllable /ng’u/ … /ng’u/.	Sikiliza: /ng’u/. Gusa silabi /ng’u/ … /ng’u/.	Syllables_11
		/ng’u/	syllables_12_sound

Items:

#	Answer Option (correct answer in gray box)				
Practice	ba	**ka**	ma	ya	za
1	ta	na	fa	**la**	ra
2	fo	**po**	ko	bo	to
3	**wa**	va	ra	ya	la
4	swu	ju	**shu**	chu	su
5	ya	vya	kwa	nja	**ja**
6	**ndi**	nzi	mbi	nji	nyi
7	ke	ge	**kwe**	twe	che
8	sa	zwa	sha	**swa**	shwa
9	be	**mbe**	nde	nge	me
10	ndwa	mbwa	ngwa	njwa	**nywa**
11	**ng’u**	dhu	gwu	nywu	ngu

4. Spelling

#	English Version	Kiswahili Version	Audio Filename
Intro Spelling	Now you will hear a word. Then you will hear the word in a sentence. Then you will hear the word again. Tap on the letters to spell the word that you hear. Tap the fruit to hear the word again. To change your answer, tap the eraser. When you finish, tap the arrow to move on. First, let's do a practice! Are you ready? Tap the arrow to begin.	Sasa utasikia neno. Kisha utalisikia neno hilo likitumika katika sentensi. Halafu utalisikia tena. Gusa herufi zenye kutamka neno unalosikia. Gusa alama ya tunda ili kulisikia tena neno hilo. Ili kubadilisha jibu lako, gusa alama ya kifutio. Ukimaliza, gusa alama ya mshale ili kuendelea. Kwanza, tufanye mazoezi! Je upo tayari? Kwa kuanza gusa alama ya mshale.	Spelling_intro
Spelling Practice	The word is water. The bucket is full of water. Water. Water.	Neno ni **maji**. Ndoo imejaa **maji. Maji. Maji.**	Spelling_practice_prompt
	That's right!	Ni sahihi!	Spelling_right
	Good try!	Umefanya vizuri!	Spelling_try
	The word maji is spelled m-a-j-i. Now you will hear a new word. Are you ready? Tap the arrow to begin.	Neno maji hutamkwa m-a-j-i. Sasa utalisikia neno jipya. Je upo tayari? Kwa kuanza gusa alama ya mshale.	Spelling_practice_answer
Spelling Help	Tap the letters to spell the word. Tap the fruit to hear the word again. To change your answer, tap the eraser. When you finish, tap the arrow to move on.	Ziguse herufi kulitamka neno. Gusa alama ya tunda ili kulisikia neno tena. Ili kubadilisha jibu lako, gusa alama ya kifutio. Ukimaliza, gusa alama ya mshale ili kuendelea.	Spelling_help
Spelling Item Prompt 1	The word is **….** [Sentence.] …. …..	Neno ni **vitu.** Sanduku lina **vitu** vingi. **Vitu. Vitu.**	Spelling_1
		vitu	spelling_1_word
2	The word is **....** [Sentence.] …. …..	Neno ni **subira. Subira** yavuta heri. **Subira. Subira.**	Spelling_2
		subira	spelling_2_word
3	The word is …. [Sentence.] …. …..	Neno ni **kubwa.** Wazazi wangu wana shamba **kubwa. Kubwa. Kubwa.**	Spelling_3
		kubwa	spelling_3_word

#	English Version	Kiswahili Version	Audio Filename
4	The word is …. [Sentence.] …. …..	Neno ni **zaidi.** Nilikula machungwa **zaidi** ya moja. **Zaidi. Zaidi.**	Spelling_4
		zaidi	spelling_4_word
5	The word is …. [Sentence.] …. …..	Neno ni **twende.** Mwalimu amesema **twende** darasani. **Twende. Twende.**	Spelling_5
		twende	spelling_5_word
6	The word is …. [Sentence.] …. …..	Neno ni **wangapi.** Darasa lenu lina wanafunzi **wangapi? Wangapi. Wangapi.**	Spelling_6
		wangapi	spelling_6_word
7	The word is …. [Sentence.] …. …..	Neno ni **shughuli.** Tunawasaidia wazazi **shughuli** za nyumbani. **Shughuli. Shughuli.**	Spelling_7
		shughuli	spelling_7_word
8	The word is …. [Sentence.] …. …..	Neno ni **hivyo.** Usiku umeingia, **hivyo**, mkalale. **Hivyo. Hivyo.**	Spelling_8
		hivyo	spelling_8_word
9	The word is …. [Sentence.] …. …..	Neno ni **mwanafunzi. Mwanafunzi** anafanya mtihani. **Mwanafunzi. Mwanafunzi.**	Spelling_9
		mwanafunzi	spelling_9_word
10	The word is …. [Sentence.] …. …..	Neno ni **yaoshwe.** Mama alisema matunda **yaoshwe**.**Yaoshwe. Yaoshwe.**	Spelling_10
		yaoshwe	spelling_10_word
11	The word is …. [Sentence.] …. …..	Neno ni **machungwa.** Watoto walienda kuchuma **machungwa. Machungwa. Machungwa.**	Spelling_11
		machungwa	spelling_11_word
12	The word is …. [Sentence.] …. …..	Neno ni **mgonjwa.** Mwalimu wetu ni **mgonjwa**. **Mgonjwa. Mgonjwa.**	Spelling_12
		mgonjwa	spelling_12_word

Items:

Practice: **maji** (water): Ndoo imejaa **maji**. (The bucket is full of water.)

1. **vitu** (things): Sanduku lina **vitu** vingi. (The bag has many things.)
2. **subira** (patience) : **Subira** yavuta heri. (Patience begets blessedness.)
3. **kubwa** (big): Wazazi wangu wana shamba **kubwa**. (My parents have a big farm.)
4. **zaidi** (more): Nilikula machungwa **zaidi** ya moja. (I ate more than one orange.)
5. **twende** (let's go): Mwalimu alisema **twende** darasani. (The teacher said let's go to class.)
6. **wangapi** (how many): Darasa lenu lina wanafunzi **wangapi**? (How many students are in your class?)
7. **shughuli** (activity): Tunawasaidia wazazi **shughuli** za nyumbani. (We help our parents in their activities (domestic chores).)
8. **hivyo** (thus): Usiku umeingia, **hivyo**, mkalale. (It's night now, thus, go to bed.)
9. **mwanafunzi** (student): **Mwanafunzi** anafanya mtihani. (The student is doing an examination.)
10. **yaoshwe** (be washed) : Mama alituambia matunda **yaoshwe**. (Mother told us to wash the fruits.)
11. **machungwa** (oranges): Watoto walienda kuchuma **machungwa**. (The children went to pick oranges.)
12. **mgonjwa** (sick person): Mwalimu wetu ni **mgonjwa**.(Our teacher is sick.)

5. Oral Reading Comprehension (shorter text)

#	English Version	Kiswahili Version	Audio Filename
Intro Practice Oral Reading	Now you will read a short story. Read the story out loud quietly and quickly. Pay attention as you read. When you finish, you will answer some questions about the story. Are you ready? Tap the arrow to begin.	Sasa utasoma hadithi fupi. Isome kwa sauti ya chini na kwa haraka. Unatakiwa kuwa makini unapoisoma. Ukimaliza kuisoma, utajibu maswali machache kuhusu hadithi hiyo. Je, upo tayari? Kwa kuanza, gusa alama ya mshale.	or_practice_intro
Oral Reading Practice	Read the story out loud. When you finish, tap the arrow to move on.	Soma hadithi kwa sauti. Ukimaliza kusoma, gusa alama ya mshale ili kuendelea.	or_practice_prompt
Intro Practice Short Comprehension	Now you will see and hear a question about the story. Tap the fruit to repeat the question or the answers. Tap your answer, then tap the arrow to move on.	Sasa utaona na kusikia swali linalohusu hadithi hiyo. Gusa alama ya tunda ili kurudia swali au majibu. Gusa jibu lako, kisha gusa alama ya mshale ili kuendelea	or_comp_practice_prompt
Practice Story	Maria walked to school in the morning. She picked a mango on the way.	Asubuhi Maria alitembea kwenda shuleni. Akiwa njiani, aliokota embe.	SHOWN ON SCREEN ONLY (no audio)
Practice Comprehension	What did Maria pick on the way to school?	Je, akiwa njiani kwenda shule, Maria aliokota nini?	or_comp_practice_question

	a. in the morning	a. asubuhi	or_comp_practice_answer_a
	b. a mango	b. embe	or_comp_practice_answer_b
	c. a banana	c. ndizi	or_comp_practice_answer_c
	d. She didn't go to school	d. hakuenda shule	or_comp_practice_answer_d
Practice Short Comprehension question	That's right!	Ni sahihi!	or_right
	Good try!	Umejaribu vizuri!	or_try
	The correct answer is "a mango."	Jibu sahihi ni 'embe'.	or_practice_answer
Intro Oral Reading	Now you will read another story. Read the story out loud quietly and quickly. Pay attention as you read. When you finish, you will answer some questions about the story. Are you ready? Tap the arrow to begin.	Sasa utasoma hadithi nyingine. Soma hadithi hiyo kwa sauti ya chini na kwa haraka. Uwe makini unapoisoma. Ukimaliza kusoma, utajibu maswali yanayohusu hadithi hiyo. Je upo tayari? Kwa kuanza, gusa alama ya mshale.	or_intro
Intro Short Comprehension	Now you will see and hear some questions about the story. Tap the fruit to repeat the question or the answers. Tap your answer, then tap the arrow to move on.	Sasa utaona na kuyasikia maswali yanayohusu hadithi uliyosoma. Gusa alama ya tunda ili kurudia swali au majibu. Gusa jibu lako, kisha gusa alama ya mshale ili kuendelea.	or_comp_intro
Short Comprehension Help	Tap the fruit to hear the question or the answers again. Tap your answer, then tap the arrow to move on.	Gusa alama ya tunda kusikia tena swali au majibu. Gusa jibu lako, kisha gusa alama ya mshale ili kuendelea.	or_comp_help

Items:

	English Translation	**Kiswahili**	**Audio Filename**
Passage	One day the dog felt hungry. He wandered around looking for food. He went to the garbage can, in the field and in the garden. He got tired. When he got to the garbage pit, he found a cat, a squirrel and a crow. The cat had four pieces of meat. There were also three bones. The dog asked the cat for some meat. He gave him two pieces of meat. The dog ate, was full and very happy. From that day on, their friendship grew stronger.	Siku moja mbwa alihisi njaa. Alizurura kutafuta chakula. Alikwenda kwenye pipa la kutupia taka, shambani na bustanini. Alichoka. Alipofika kwenye shimo la taka, aliwakuta paka, kicheche na kunguru. Paka alikuwa na vipande vinne vya minofu. Pia, kulikuwa na mifupa mitatu. Mbwa alimuomba paka mnofu mmoja. Akampa minofu miwili. Mbwa alikula, akashiba na akafurahi sana. Tangu siku hiyo urafiki wao uliimarika.	ON SCREEN ONLY (no audio)
Question 1	1. At the beginning of the story, what did the dog feel?	1. Mwanzoni mwa hadithi, mbwa alihisi nini?	or_q1
Q1 Answer Options	a. happiness	a. furaha	or_q1a
	b. fatigue	b. uchovu	or_q1b
	c. friendship	c. urafiki	or_q1c
	***d. hunger**	d. njaa	or_q1d
Question 2	2. Where did the dog get food?	2. Mbwa alipata chakula wapi?	or_q2
Q2 Answer Options	a. behind the house	a. nyuma ya nyumba	or_q2a
	***b. in the garbage pit**	b. kweye shimo la taka	or_q2b
	c. in the field	c. shambani	or_q2c
	d. in the garden	d. bustanini	or_q2d
Question 3	3. Who did the dog ask for food?	3. Mbwa alimuomba nani chakula?	or_q3
Q3 Answer Options	***a. cat**	a. paka	or_q3a
	b. squirrel	b. kicheche	or_q3b
	c. crow	c. kunguru	or_q3c
	d. another dog	d. mbwa mwingine	or_q3d
Question 4	4. How many fillets did the cat give the dog?	4. Paka alimpa mbwa minofu mingapi?	or_q4
Q4 Answer Options	a. one	a. mmoja	or_q4a
	***b. two**	b. miwili	or_q4b
	c. three	c. mitatu	or_q4c
	d. four	d. minne	or_q4d
Question 5	5. What behavior did the cat show towards the dog?	5. Paka alionesha tabia gani kwa mbwa?	or_q5
Q5 Answer Options	a. selfish behavior	a. tabia ya uroho	or_q5a
	b. greedy behavior	b. tabia ya uchoyo	or_q5b
	***c. loving behavior**	c. tabia ya upendo	or_q5c
	d. dishonest behavior	d. tabia ya uongo	or_q5d

	English Translation	Kiswahili	Audio Filename
Question 6	6. How did the cat and dog strengthen their friendship?	6. Kwa nini paka na mbwa waliimarisha urafiki wao?	or_q6
Q6 Answer Options	a. Because they met in a garbage dump.	a. Kwa sababu walikutana kwenye shimo la taka.	or_q6a
	***b. Because the cat helped the dog with food.**	b. Kwa sababu paka alimsaidia mbwa chakula.	or_q6b
	c. Because the dog needed friends.	c. Kwa sababu mbwa alikuwa anahitaji marafiki.	or_q6c
	d. Because they all live together.	d. Kwa sababu wote wanaishi pamoja.	or_q6d

6. Silent Reading Comprehension (longer text)

	English Version	Kiswahili Version	Audio Filename
Intro Silent Reading	Now you will read another story. This time read the story silently to yourself in your head. Pay attention as you read. When you finish, you will answer some questions about the story. Are you ready? Tap the arrow to begin.	Sasa utasoma hadithi nyingine. Kwa sasa, utasoma kimya kimya. Uwe makini unaposoma. Ukimaliza kusoma, utajibu maswali yanayohusu hadithi hiyo. Je upo tayari? Kwa kuanza gusa alama ya mshale.	sr_intro
Silent Reading Help	Read the story silently in your head. When you finish, tap the arrow to move on.	Soma hadithi hii kimya kimya. Ukimaliza, gusa alama ya mshale ili kuendelea.	sr_help
Intro Long Comprehension	Now you will see and hear some questions about the story. Tap the left arrow to see the story again. Tap the fruit to repeat the question or the answers. Tap your answer, then tap the right arrow to move on.	Sasa utayaona na kuyasikia baadhi ya maswali yanayohusu hadithi hiyo. Gusa alama ya mshale wa kushoto ili uione tena hadithi. Ili kurudia swali au majibu, gusa alama ya tunda. Gusa jibu lako, kisha gusa alama ya mshale wa kulia ili kuendelea.	sr_comp_intro
Long Comprehension Help	Tap the left arrow to see the story again. Tap the fruit to hear the question or the answers again. Tap your answer, then tap the right arrow to move on.	Ili uione tena hadithi, gusa alama ya mshale wa kushoto. Gusa alama ya tunda kusikia tena swali au majibu. Gusa jibu lako, kisha gusa alama ya mshale wa kulia ili kuendelea.	sr_comp_help

Items:

	English Translation	**Kiswahili Version**	**Audio Filename**
Passage	Kibibi lived with her mother in the village. One day her mother suddenly collapsed and fainted. Kibibi screamed for help, and the neighbors ran to help her. When they arrived at the hospital, the doctor and nurses gave her urgent treatment. After two days of examination, her condition improved and she was allowed to return home. Another day, a teacher was bitten by a snake on the leg. Without hesitation, a student tore off a piece of cloth and tied it over the wound to prevent the poison from spreading. As the doctor passed by, he found the student taking care of the teacher. The doctor helped clean the wound with cotton, water, and medicine and bandaged it. Then they headed to the hospital. As they continued their journey, the teacher's condition worsened, which greatly frightened the student. However, because the doctor and the student had helped him earlier, the teacher was also able to recover in the hospital. Kibibi was very happy to see her mother and the teacher recover, which made her want to become a doctor.	Kibibi aliishi na mama yake kijijini. Siku moja mama yake alianguka ghafla na kuzimia. Kibibi alipiga kelele kuomba msaada, na majirani wakamkimbilia kumsaidia. Walipofika hospitali, daktari na wauguzi walimpa matibabu ya haraka. Baada ya siku mbili za uchunguzi, hali yake iliimarika na akaruhusiwa kurudi nyumbani. Siku nyingine, mwalimu aliumwa na nyoka kwenye mguu. Bila kusita, mwanafunzi alirarua kipande cha kanga na kukifunga juu ya jeraha ili sumu isisambae. Daktari alipokuwa akipita alimkuta mwanafunzi akimhudumia mwalimu. Daktari alisaidia kusafisha kidonda kwa pamba, maji, na dawa na kuifunga. Kisha wakaelekea hospitali. Wakati wakiendelea na safari, hali ya mwalimu ilizidi kuwa mbaya, jambo ambalo lilimtia hofu sana mwanafunzi. Hata hivyo, kwa sababu daktari na mwanafunzi walikuwa wamemsaidia mapema, mwalimu huyo pia aliweza kupata nafuu hospitalini. Kibibi alifurahi sana kupata kupona kwa mama yake na mwalimu, jambo ambalo lilimfanya atamani kuwa daktari.	ON SCREEN ONLY (no audio)
Question 1	1. Who fell and fainted?	1. Nani alianguka na kuzimia?	sr_q1
Q1 Answer Options	a. doctor	a. daktari	sr_q1a
	b. a student	b. mwanafunzi	sr_q1b
	***c. mother**	c. mama	sr_q1c
	d. teacher	d. mwalimu	sr_q1d
Question 2	2. When her mother fell and fainted, where was she taken?	2. Mama yake kibibi alipoanguka na kuzimia alipelekwa wapi?	sr_q2

	English Translation	Kiswahili Version	Audio Filename
Q2 Answer Options	a. She was sent on her way.	a. Alipelekwa njiani.	sr_q2a
	***b. She was taken to the hospital.**	b. Alipelekwa hospitalini.	sr_q2b
	c. She was taken to the village.	c. Alipelekwa kijijini.	sr_q2c
	d. She was sent to a neighbor.	d. Alipelekwa kwa jirani.	sr_q2d
Question 3	3. After two days of treatment, what happened to Kibibi's mother?	3. Baada ya siku mbili za matibabu, nini kilitokea kwa mama yake Kibibi?	sr_q3
Q3 Answer Options	a. She worsened.	a. Alizidiwa.	sr_q3a
	b. She fell.	b. Alianguka.	sr_q3b
	c. She fainted.	c. Alizimia.	sr_q3c
	***d. She recovered.**	d. Alipona.	sr_q3d
Question 4	4. What did the student use to help the teacher who was bitten by a snake?	4. Mwanafunzi alitumia nini kumsaidia mwalimu aliyegongwa na nyoka?	sr_q4
Q4 Answer Options	***a. Piece of the cloth**	a. kanga	sr_q4a
	b. Bandage	b. bandeji	sr_q4b
	c. Cotton	c. pamba	sr_q4c
	d. Medicine	d. dawa	sr_q4d
Question 5	5. Who helped the student take care of the teacher?	5. Ni nani aliyemsaidia mwanafunzi kumhudumia mwalimu aliyegongwa na nyoka?	sr_q5
Q5 Answer Options	a. nurse	a. muuguzi	sr_q5a
	b. another teacher	b. mwalimu mwingine	sr_q5b
	***c. doctor**	c. daktari	sr_q5c
	d. neighbor	d. jirani	sr_q5d
Question 6	6. Why did Kibibi scream?	6. Kwa nini Kibibi alipiga yowe?	sr_q6
Q6 Answer Options	***a. Because her mom fell and fainted.**	a. Kwa sababu mama yake alianguka na kuzimia.	sr_q6a
	b. Because her mother lost her life.	b. Kwa sababu mama yake alipoteza maisha.	sr_q6b
	c. Because her mother was bitten by a snake.	c. Kwa sababu mama yake aligonwa na nyoka.	sr_q6c
	d. Because her teacher was bitten by a snake.	d. Kwa sababu mwalimu wake aligonwa na nyoka.	sr_q6d
Question 7	7. In this story, who did Kibibi want to be?	7. Kwenye hadhithi hii, Kibibi alipenda kuwa nani?	sr_q7
Q7 Answer Options	a. mother	a. mama	sr_q7a
	b. teacher	b. mwalimu	sr_q7b
	c. nurse	c. muuguzi	sr_q7c
	***d. doctor**	d. daktari	sr_q7d
Question 8	8. What made Kibibi want to become a doctor?	8. Ni nini kilichomfanya Kibibi atamani kuwa daktari?	sr_q8
Q8	***a. Because the mother and the teacher were cured.**	a. Kwa sababu mama na mwalimu walipona.	sr_q8a

	English Translation	Kiswahili Version	Audio Filename
Answer Options	b. Because the student survived.	b. Kwa sababu mwanafunzi alipona.	sr_q8b
	c. Because they had a big hospital.	c. Kwa sababu walikuwa na hospitali kubwa.	sr_q8c
	d. Because there were so many snakes.	d. Kwa sababu nyoka walikuwa wengi.	sr_q8d

7. Vocabulary

#	English Prompt	Kiswahili Version	Audio Filename
Intro Vocab	Now we will play a game with words! You will see and hear four words like this. Some of the words are easier and some are harder. Tap the word that you know the best. First, let's do a practice! Are you ready? Tap the arrow to begin.	Sasa tutacheza mchezo kutumia maneno! Utayaona na kuyasikia maneno manne kama haya. Baadhi ya maneno ni rahisi na mengine ni magumu. Gusa neno ambalo unalijua vizuri zaidi. Kwanza, tuanze na mazoezi! Je upo tayari? Kwa kuanza, gusa alama ya mshale.	vocab_intro
Practice Vocab	Tap the word that you know the best.	Gusa neno ambalo unalijua vizuri zaidi.	vocab_practice_prompt
	Most children probably know the meaning of the word **'jibu'**. It is the easiest word here. All the other words are harder. Now you will see some more words. Tap the word that you know the best, the word that is easiest for you. Are you ready? Tap the arrow to begin.	Watoto wengi huenda wanajua maana ya neno **'jibu'**. Hilo ndilo neno rahisi zaidi hapa. Maneno mengine yote ni magumu zaidi. Sasa utayaona maneno mengine. Gusa neno ambalo unalijua zaidi, neno lililo rahisi zaidi kwako. Je upo tayari? Gusa alama ya mshale kuanza	vocab_practice_answer
Vocab Item Prompt	Tap the word that you know the best.	Gusa neno unalolijua zaidi.	vocab_item_prompt
Vocab Help	Tap the word that you know the best. Then tap the arrow.	Gusa neno ambalo unalijua zaidi. Kisha gusa alama ya mshale.	vocab_help

Items:

						English (not shown on screen!)
Practice		muza	vivi	jibu	kewa	answer
Std 1	1	foroko	pimapi	jokofu	dikiha	refrigerator
	2	**likoyo**	**kurunzi**	**ngakomi**	**kosoni**	torch
	3	**bitu**	**ngiwe**	**kinywa**	**tevyo**	mouth (scientific)
	4	**fyekeo**	**ganu**	**tala**	**rupe**	slasher
	5	**pwekeru**	**tipene**	**mulagu**	**gwaride**	parade
	6	**jitihada**	**tumataju**	**bisuriwa**	**awikaha**	effort
Std 2	7	**nzocha**	**goge**	**chanzo**	**shoke**	source
	8	**buta**	**mbeje**	**pifu**	**bunge**	parliament
	9	riga	naji	ngoja	lesho	wait
	10	**hakika**	**rishadi**	**mbetoki**	**zarobi**	certainly
	11	**kipopwi**	**njenga**	**tibu**	**lukuti**	treat (verb)
	12	**kamula**	**kadri**	**pwiliko**	**twarama**	as much as
	13	**rakinja**	**sambata**	**fekero**	**chagua**	select
	14	**tabasamu**	**dheluji**	**lomwisaku**	**jithelu**	grin
Std 3	15	**njungupawe**	**poku**	**tonzi**	**dira**	compass
	16	**tibuku**	**ngunjunawe**	**serereka**	**bimakima**	sliding
	17	theluji	tongoka	ruda	hekeresha	snow
	18	**ngaje**	**jenga**	**thekaneke**	**lengeke**	construct
	19	kipupwe	nganjulapo	mungune	rufishi	winter
	20	nyemung'u	selugi	nganye	tenzi	poem

8. Syntax

#	English Prompt	Kiswahili Version	Audio Filename
Intro Syntax	Now you will hear four sentences. Listen carefully. One of the sentences is good. The other sentences are bad – they are not true or they don't make any sense. Here is an example. Tap the good sentence.	Sasa utasikiliza sentensi nne. Sikiliza kwa makini. Moja ya sentensi hizo ni sahihi na inaleta maana. Sentensi zingine siyo sahihi – zimekosewa na hazina maana . Hapa kuna mfano. Gusa sentensi sahihisahihi.	**syntax_intro**
Practice Syntax Item	a. Juma cries a shoe.	a. Juma analia kiatu.	syntax_practice_q00a
	b. Juma cries rice.	b. Juma analia wali.	syntax_practice_q00b
	c. *Juma cries loudly.	c. Juma analia kwa sauti.	syntax_practice_q00c
	d. Juma cries a bottle.	d. Juma analia chupa	syntax_practice_q00d
Practice Syntax Answer	The correct answer is Sentence C, **Juma cries loudly**. Sentence C is a good sentence because it makes sense! The other sentences don't make sense in Kiswahili; they are bad sentences. Now you will hear some more sentences. Are you ready? Tap the arrow to begin.	Jibu sahihi ni Sentensi C, Juma analia kwa sauti. Sentensi C ni sentensi sahihi kwa sababu ni yenye maana! Sentensi zingine hazina maana katika Kiswahili; siyo sentensi sahihi. Sasa utasikia sentensi nyingine. Je upo tayari? Kwa kuanza, gusa alama ya mshale.	syntax_practice_answer
Syntax Item Prompt	Tap the good sentence.	Gusa sentensi sahihi.	syntax_practice_prompt
Syntax Help	Tap the good sentence. Tap the fruit to hear the sentence again. When you finish, tap the arrow to move on.	Gusa sentensi sahihi. Gusa alama ya tunda kuisikiliza tena sentensi . Ukimaliza, gusa alama ya mshale ili kuendelea.	syntax_help

Items:

Target grammatical structure	English translation	Kiswahili version	Audio Filename
1. Sentence word order	**a. *The house has a door.**	a. Nyumba ina mlango.	
	b. A door has a house.	b. Mlango una nyumba.	
	c. A door house has.	c. Mlango nyumba ina.	
	d. A door house has.	d. Mlango nyumba una.	

2. If-conditional clause, word order	a. If you exam hard, you will pass the study.	a. Kama mtihani kwa bidii, utafaulu ukisoma.	
	b. If you pass the exam, you will study very hard.	b. Kama utafaulu mtihani, ukisoma kwa bidi.	
	c. *If you study hard, you will pass the exam.	c. Kama ukisoma kwa bidii, utafaulu mtihani.	
	d. If you hard exam, you will pass the study.	d. Kama kwa bidi mtihani, utafaulu ukisoma.	
3. Verb tense	**a. *You were watching TV yesterday.**	a. Jana ulikuwa unaangalia televisheni.	
	b. You are watching TV yesterday.	b. Jana unaangalia televisheni.	
	c. You will be watching TV yesterday.	c. Jana utakuwa unaangalia televisheni.	
	d. You might be watching TV yesterday.	d. Jana huenda utakuwa unaangalia televisheni.	
4. Locative/word order	a. The child the market went.	a. Mtoto sokoni alikwenda.	
	b. Went the child the market.	b. Alikwenda mtoto sokoni.	
	c. * The child went to the market.	c. Mtoto alikwenda sokoni.	
	d. The market the child went.	d. Sokoni mtoto alikwenda.	
5. Relative clauses	a. He is caught that the thief escaped.	a. Amekamatwa aliyetoroka mwizi.	
	b. *The thief that escaped has been caught.	b. Mwizi ambaye alitoroka amekamatwa.	
	c. That the thief has been caught escaped.	c. Ambaye mwizi amekamatwa alitoroka.	
	d. He escaped the thief that has been caught.	d. Aliyetoroka mwizi ambaye amekamatwa.	
6. Demonstrative agreement with noun class	a. These child is notorious.	a. Mtoto hawa ni mtundu.	
	b. Those child is notorious.	b. Mtoto wale ni mtundu.	
	c. That child are notorious.	c. Mtoto yule ni watundu.	
	d. *This child is notorious.	d. Mtoto huyu ni mtundu.	

7. Possessive agreement with noun class	a. ***His/her book is torn.**	a. Kitabu chake kimechanika.	
	b. His/her (singular) book are torn.	b. Kitabu vyake vimechanika.	
	c. Their book are torn.	c. Kitabu vyao vimechanika.	
	d. Their book is torn. (interchanged noun classes).	d. Kitabu yao imechanika.	
8. Infinitive of purpose	a. People get firewood to cut down trees.	a. Watu hupata kuni ili kukata miti.	
	b. Firewood cuts people to get trees down.	b. Kuni hukata watu ili kupata miti.	
	c. *People cut down trees to get firewood.	c. Watu hukata miti ili kupata kuni.	
	d. Trees get people to cut firewood.	d. Miti hupata watu ili kukata kuni.	

EGMA

1. Introduction

#	English Prompt	Kiswahili	
Beginning Prompt	Are you ready? Tap the arrow to begin.	Je upo tayari? Kwa kuanza, gusa alama ya mshale.	Begining_prompt
Paper Aside	Please take your paper and pencil and place them to the side. Only use them when instructed to do so.	Tafadhali chukua karatasi na kalamu yako na kuviweka pembeni. Tumia karatasi na kalamu unapoelekezwa kufanya hivyo.	paper_aside
Practice Intro	For the next 30 seconds, use the number buttons at the top of the screen to enter whatever you like. You can also use the eraser to erase a number or letter.	Kwa sekunde 30 zijazo, tumia alama ya vitufe vya namba vilivyoko juu ya skrini kuingiza alama au herufi unayotaka. Pia unaweza kutumia alama ya kifutio kufuta namba au herufi.	Practice_intro

2. Number Identification

#	English Prompt	Kiswahili	
Number Identification Intro	Tap the Fruit to hear it say a number aloud. Use the digits 0 to 9 at the top to write the number you hear. To change your answer, tap the eraser. When you finish, tap the arrow to move on. Now, you give it a try.	Gusa alama ya tunda ili litaje namba kwa sauti. Tumia tarakimu 0 hadi 9 zilizo juu kuandika namba unayoisikia. Ili kulibadilisha jibu lako, gusa alama ya kifutio. Ukimaliza, gusa alama ya mshale ili kuendelea. Sasa, na wewe jaribu.	Number_id_intro
Number Identification Help	Tap the fruit to hear it say a number aloud. Use the digits 0 to 9 at the top to write the number you hear. To change your answer, tap the eraser. When you finish, tap the arrow to move on.	Gusa alama ya tunda ili litaje namba kwa sauti. Tumia tarakimu 0 hadi 9 zilizo juu kuandika namba unayosikia. Ili kulibadilisha jibu lako, gusa alama ya kifutio. Ukimaliza, gusa alama ya mshale ili kuendelea.	Number_id_help
1	one	Moja	num_id1
2	two	Mbili	num_id2
3	nine	Tisa	num_id3

4	zero	Sifuri	num_id4
5	twelve	Kumi na mbili	num_id5
6	forty-five	Arobaini na tano	num_id6
7	thirty-nine	Thelathini na tisa	num_id7
8	eighty	Themanini	num_id8
9	seventy-four	Sabini na nne	num_id9
10	sixty-six	Sitini na sita	num_id10
11	one hundred and eight	Mia moja na nane	num_id11
12	five hundred and eighty-seven	Mia tano na themanini na saba	num_id12
13	nine hundred and eighty-nine	Mia tisa na themanini na tisa	num_id13

3. Number Comparison

#	English Prompt	Kiswahili	
Number Comparison Intro	Tap the biggest number. To change your answer, tap the number you want to choose. When you finish, tap the arrow to move on.	Gusa namba kubwa zaidi kuliko zote. Ili kubadilisha jibu lako, gusa namba unayotaka kuchagua. Ukimaliza, gusa mshale ili kuendelea.	Num_comp_intro
Number Comparison Item Prompt	Tap the biggest number.	Gusa namba kubwa zaidi kuliko zote.	Num_comp_prompt
Number Comparison Help	Tap the biggest number. To change your answer, tap the number you want to choose. When you finish, tap the arrow to move on.	Gusa namba kubwa zaidi kuliko zote. Ili kubadilisha jibu lako, gusa namba unayotaka kuchagua. Ukimaliza, gusa alama ya mshale ili kuendelea.	Num_comp_help
1	7, 5, 9		num_comp_1
2	11, 24, 13		num_comp_2
3	39, 23, 26		num_comp_3
4	49, 47, 56		num_comp_4
5	65, 67, 64		num_comp_5
6	94, 78, 69		num_comp_6
7	146, 137, 153		num_comp_7
8	287, 534, 396		num_comp_8
9	632, 623, 626		num_comp_9
10	867, 765, 964		num_comp_10

4. Missing Number

#	English Prompt	Kiswahili	
Missing Number Intro	Here are some numbers. What number is missing?	Hizi ni baadhi ya namba. Ni namba gani inayokosekana? Tumia	Miss_num_intro

	Use the digits 0 to 9 at the top to write the missing number.	tarakimu 0 hadi 9 zinazoonekana juu ili kujaza namba inayokosekana.	
Missing Number Practice 1	Now you give it a try. Here are some numbers: 1, 2, and 4. What number is missing?	Sasa jaribu. Hizi hapa ni baadhi ya namba: 1, 2, _ 4. Ni namba gani inayokosekana?	miss_num_practice_01
Missing Number Practice 2	Here are some numbers: 5, 10, and 15. What number is missing?	Hizi hapa ni baadhi ya namba: 5, 10, _ 15. Ni namba gani inayokosekana?	miss_num_practice_02
Missing Number Item Prompt	What number is missing?	Ni namba gani inayokosekana?	Miss_num_prompt
Missing Number Help	Use the digits 0 to 9 at the top to write the missing number. To change your answer, tap the eraser.	Tumia tarakimu 0 hadi 9 zinazoonekana juu ili kujaza namba inayokosekana. Ili kubadilisha jibu lako, gusa alama ya kifutio.	Miss_num_help
1	5, 6, 7, [ ]		miss_num_01
2	14, 15, [ ], 17		miss_num_02
3	20, [ ], 40, 50		miss_num_03
4	[ ], 300, 400, 500		miss_num_04
5	2, 4, 6, [ ]		miss_num_05
6	348, 349, [ ], 351		miss_num_06
7	28, [ ], 24, 22		miss_num_07
8	30, 35, [ ], 45		miss_num_08
9	550, 540, 530, [ ]		miss_num_09
10	3, 8, [ ], 18		miss_num_10

5. Addition 1

#	English Prompt	Kiswahili	
Addition Intro	Here is an addition problem. Use the digits 0 to 9 at the top to write the answer to the problem. Now, you give it a try.	Hapa kuna maswali ya kujumlisha. Tumia tarakimu 0 hadi 9 zinazoonekana juu ili kuandika jibu la swali. Hebu jaribu sasa.	I1_addition_intro
Addition Item Prompt	Use the digits 0-9 at the top to write the answer to the problem.	Tumia tarakimu 0 hadi 9 zinazoonekana juu kuandika jibu la swali.	I1_addition_prompt
Addition Help	Here is an addition problem. Use the digits 0 to 9 at the top to write the	Hapa kuna maswali ya kujumlisha. Tumia tarakimu 0 hadi 9 zinazoonekana juu kuandika jibu la swali.	I1_addition_help

	answer to the problem.		
1	1 + 3 =		I1_addition_01
2	6 + 2 =		I1_addition_02
3	3 + 3 =		I1_addition_03
4	8 + 1 =		I1_addition_04
5	2 + 8 =		I1_addition_05
6	9 + 8 =		I1_addition_06
7	10 + 2 =		I1_addition_07
8	8 + 10 =		I1_addition_08
Addition Transition	Let's continue with some more problems. You may use paper and pencil if you want to. You do not have to do so.	Tuendelee na baadhi ya maswali mengine. Unaweza kutumia karatasi na kalamu ikiwa unataka. Siyo lazima kufanya hivyo.	I1_addition_trans
1	13 + 6 =		I2_addition_01
2	18 + 7 =		I2_addition_02
3	12 + 14 =		I2_addition_03
4	22 + 37 =		I2_addition_04
5	38 + 26 =		I2_addition_05

6. Subtraction 1

#	English Prompt	Kiswahili	
Subtraction Intro	Here is a subtraction problem. Use the digits 0 to 9 at the top to write the answer to the problem.	Hapa ni maswali ya kutoa. Tumia tarakimu 0 hadi 9 zinazoonekana juu kuandika jibu la swali .	I1_subtraction_intro
1	4 – 3 =		I1_subtraction_01
2	8 – 2 =		I1_subtraction_02
3	6 - 3 =		I1_subtraction_03
4	9 – 1 =		I1_subtraction_04
5	10 – 8 =		I1_subtraction_05
6	17 – 8 =		I1_subtraction_06
7	12 – 2 =		I1_subtraction_07
8	18 – 10 =		I1_subtraction_08
Subtraction Transition	Next are a few more problems for you to solve. You may use paper and pencil if you want to. You do not have to do so.	Yafuatayo ni maswali zaidi ya kujibu. Unaweza kutumia karatasi na kalamu ikiwa unataka. Siyo lazima kufanya hivyo.	I1_subtraction_trans
1	19 – 6 =		I2_subtraction_01
2	25 – 7 =		I2_subtraction_02
3	26 – 14 =		I2_subtraction_03
4	59 – 37 =		I2_subtraction_04
5	64 – 26 =		I2_subtraction_05

7. Word Problems

#	English Prompt	Kiswahili	
Word Problems Intro	Here are some problems for you to solve. Tap the Fruit to hear it say the problem aloud. Tap the Fruit to hear the problem again. You may use pencil, paper, and counters if you want to. You don't have to use them. Now, you give it a try.	Haya hapa ni baadhi ya maswali unayotakiwa kuyajibu. Gusa alama ya tunda ili usikie swali likisomwa kwa sauti. Gusa tena alama ya tunda ili ulisikie swali hilo. Unaweza kutumia kalamu, karatasi, na vihesabio ikiwa unataka. Lakini siyo lazima uvitumie vitu hivyo. Jaribu sasa.	word_problem_intro
Word Problems Practice	There are three children in the bus. One child gets off the bus. How many children are left on the bus?	Ndani ya basi kuna watoto watatu. Mtoto mmoja ameshuka kutoka kwenye basi. Je, ni watoto wangapi wamebaki ndani ya basi?	word_problem_practice
	The correct answer is two. If there are three children on the bus and one child gets off, there are two children left on the bus.	Jibu sahihi ni watoto wawili. Ikiwa kuna watoto watatu ndani ya basi na mtoto mmoja ameshuka, hivyo watoto wawili watabaki ndani ya basi.	word_problem_practice_ans
Word Problems Help	Tap the fruit to hear it say a problem aloud. You may use pencil, paper, and counters if you want to. You do not have to use them.	Gusa alama ya tunda ili ulisikie swali likisomwa kwa sauti. Unaweza kutumia kalamu, karatasi, na vihesabio ikiwa unataka. Lakini, siyo lazima uvitumie vitu hivyo.	word_problem_help
1	There are 2 children in the bus. 3 more children get on the bus. How many children are there on the bus altogether?	Kuna watoto 2 ndani ya basi. Watoto 3 zaidi wanapanda kwenye basi. Jumla kutakuwa na watoto wangapi ndani ya basi?	word_problem_01
2	There are 6 children in the bus. 2 of those children are boys, and all the rest are girls. How many girls are in the bus?	Kuna watoto 6 ndani basi. Kati ya hao, 2 ni wavulana, na waliobaki wote ni wasichana. Je kutakuwa na wasichana wangapi ndani ya basi?	word_problem_02
3	There are 2 oranges in [Juma's basket. There are 8 oranges in [Mariamu]'s basket.	Kuna machungwa 2 kwenye kikapu cha Juma. Pia, kuna machungwa 8 kwenye kikapu cha	word_problem_03

	How many more oranges must be added to [Juma]'s basket so that it has the same number of oranges as [Mariamu's basket?	Mariamu. Je ni machungwa mangapi yanapaswa kuongezwa kwenye kikapu cha Juma ili yafikie idadi sawa ya machungwa na iliyopo kwenye kikapu cha Mariamu.	
4	There is a bus with some children. 2 more children get in the bus. Now there are 9 children in the bus. How many children were in the bus to begin with?	Kuna basi lenye watoto kadhaa. Watoto 2 zaidi wanapanda kwenye basi. Sasa kuna jumla ya watoto 9 ndani ya basi. Je, ni watoto wangapi walikuwapo awali ndani ya basi?	word_problem_04
5	There are 12 candies. 4 children share the candies equally. How many candies does each child get?	Kuna peremende 12. Watoto 4 wanagawana peremende hizo kwa usawa. Je kila mtoto atapata peremende ngapi?	word_problem_05
6	There are 5 seats on the van. There are 2 children on each seat. How many children are on the van altogether?	Kuna viti 5 kwenye gari dogo. Kila kiti wameketi watoto 2. Je, jumla kuna watoto wangapi waliopo ndani ya gari hilo?	word_problem_06

8. Transitions

	English Prompt	Kiswahili Version	
Bubble Game Transition	Now let's take a short break and play a game. You will see some bubbles appear on the screen. Every time you see a bubble, tap it as fast as you can. Ready? Tap the arrow to begin.	Sasa tupate mapumziko mafupi na tucheze mchezo. Utaona maputo yakijitokeza kwenye skrini. Kila unapoona puto, gusa kwa haraka kadri uwezavyo. Je upo tayari? Kwa kuanza, gusa alama ya mshale.	`math_76-bubble-game-intro`
Transition 1	Good job! Now let's do something different.	Kazi nzuri! Sasa tufanye kitu tofauti.	`Transition_1`

Transition 2	Now let's move on to the next activity.	Sasa tuendelee na shughuli inayofuata.	Transition_2
Transition 3	Good job! Let's do one last activity.	Kazi nzuri! Hebu tufanye shughuli moja ya mwisho.	Transition_3
Return Tablet	You finished! You did a great job! Thank you for playing our game. Please return the tablet to the teacher who gave it to you.	Umemaliza! Umefanya kazi nzuri! Asante kwa kucheza mchezo wetu. Tafadhali rudisha kishikwambi kwa mwalimu aliyekupatia	return_tablet